

printf(): It is a predefined function available in stdio.h

it is the major output function in c.

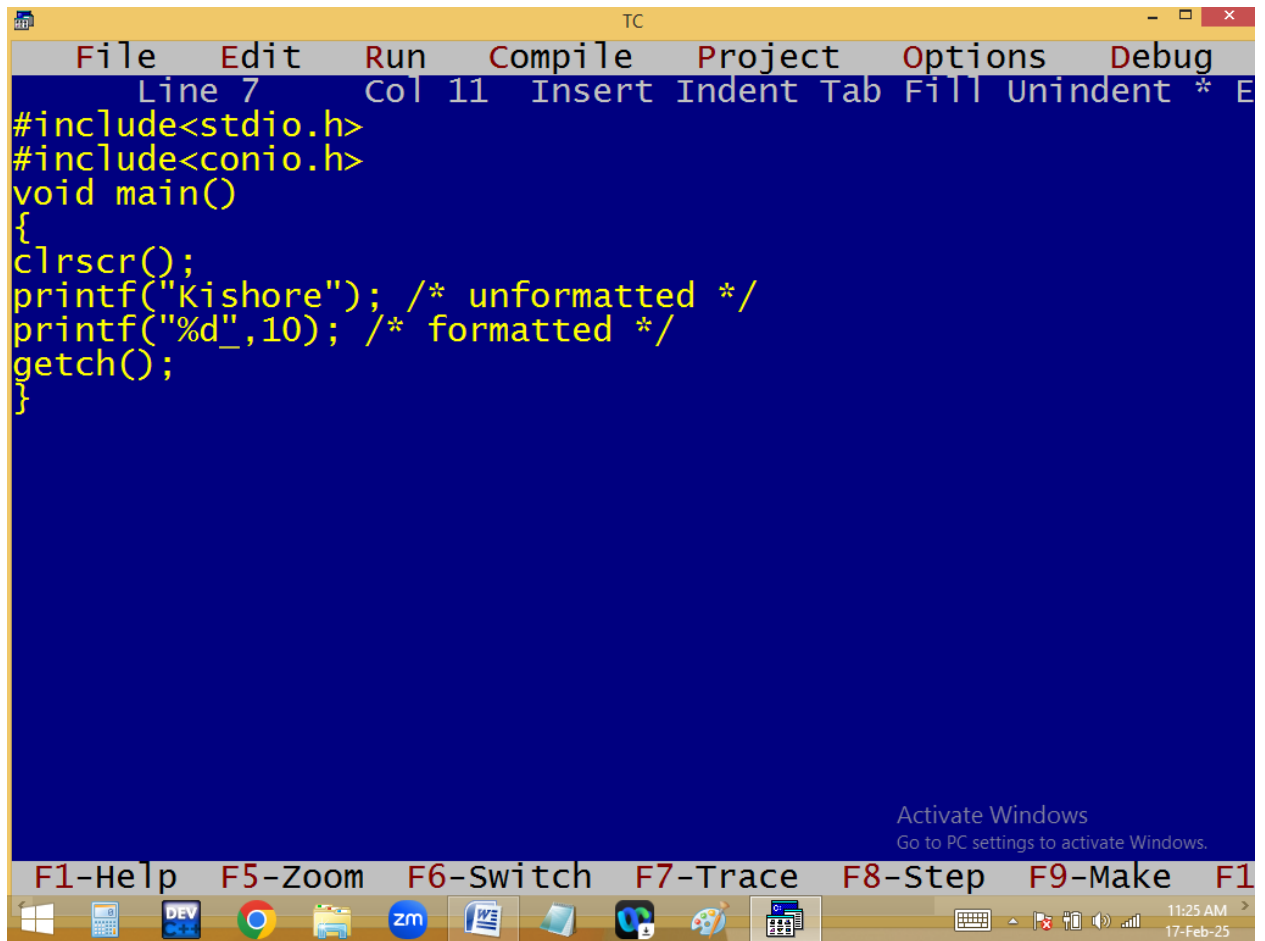
in printf, f means formatted.

Printf always refers **standard output device** i.e. monitor.

Note:

1. Printf always return an int that indicates the no of printable characters.

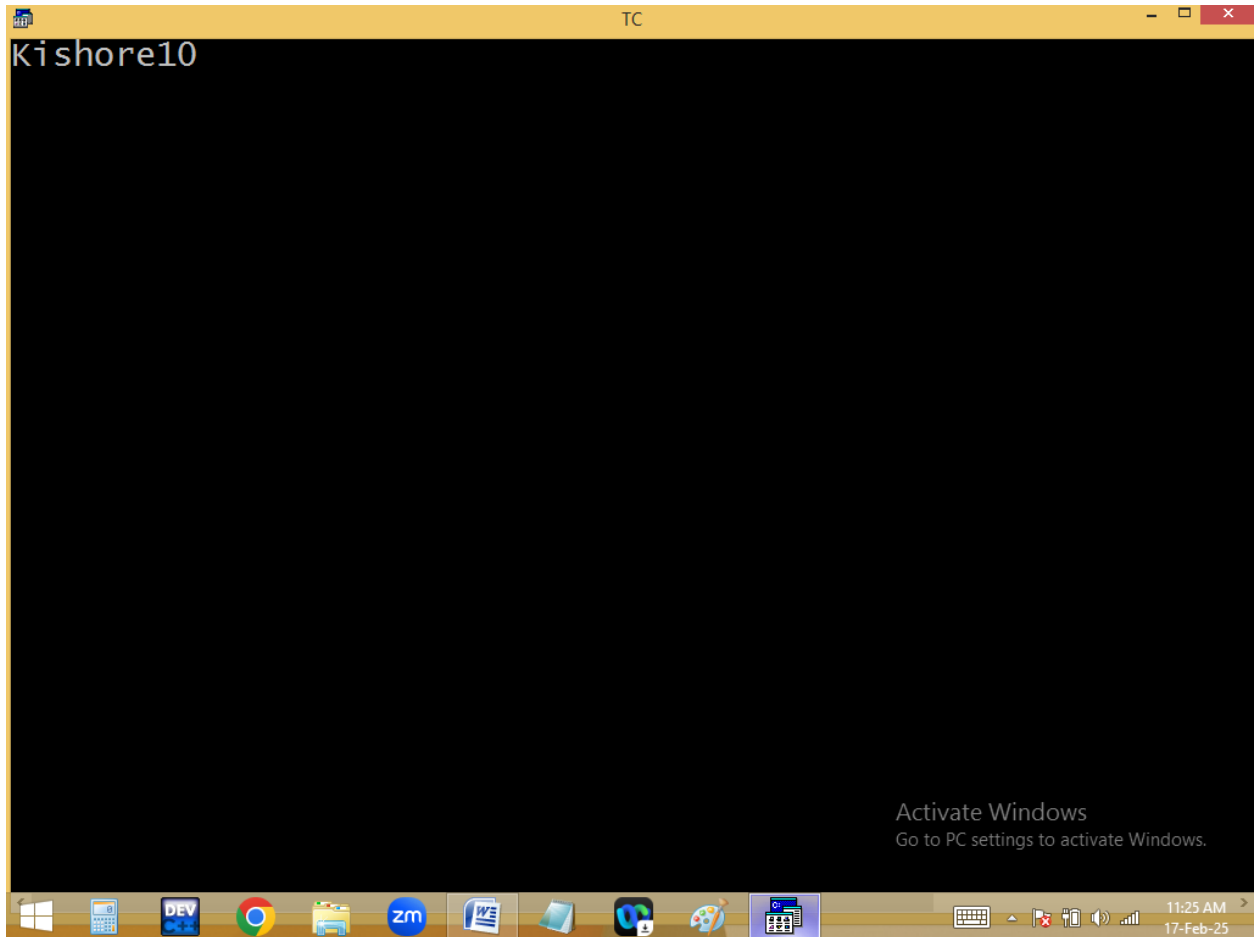
2. Printf can perform both formatted and unformatted outputs.



The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 7' and 'Col 11'. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore"); /* unformatted */
printf("%d",10); /* formatted */
getch();
}
```

At the bottom of the window, there is an 'Activate Windows' watermark with the text 'Go to PC settings to activate Windows.' and a taskbar with various application icons. The taskbar also displays the time '11:25 AM' and the date '17-Feb-25'.



3. In printf execution order is right to left and printing is left to right.
4. In printf the first argument should be in “ ”.
5. In printf everything printed as it is except conversion characters and back slash characters.

Syntax:

```
int printf( " [text] [ conversion characters / format specifiers / format  
string ] " [ , variables ] [ , expressions ] );
```

```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 16 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
float b=1.2;
clrscr();
printf("Hi\n"); /* unformatted */
printf("a=%d\n",a); /* formatted */
printf("b=%f\n",b);
printf("a=%d, b=%f\n",a,b);
printf("Sum=%f\n",a+b);
printf("a=%d, b=%f, sum=%f\n",a,b,a+b);
printf("%d + %f = %f\n",a,b,a+b);
printf("%d",a);_
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

11:36 AM
17-Feb-25

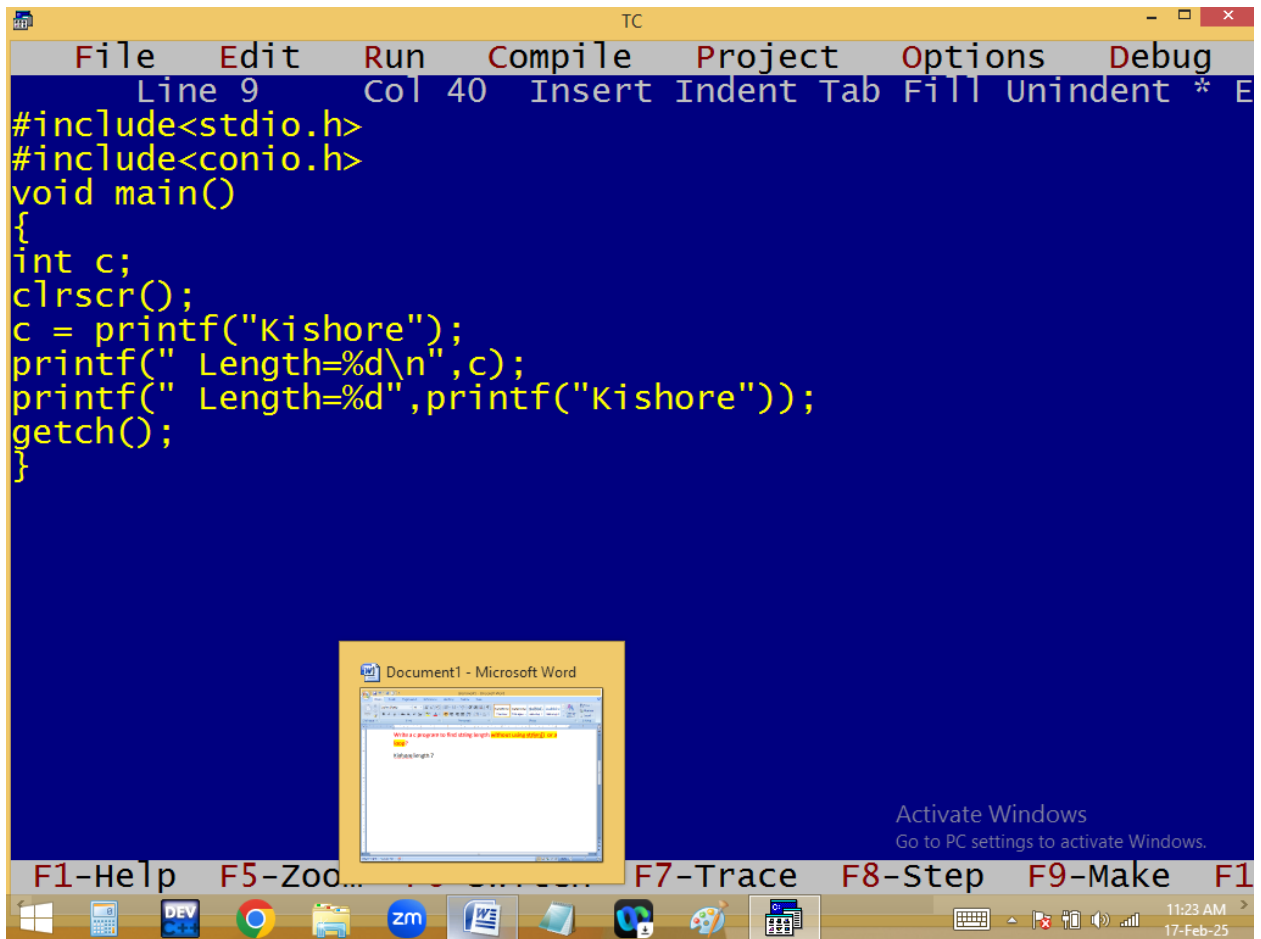
```
TC
Hi
a=10
b=1.200000
a=10, b=1.200000
Sum=11.200000
a=10, b=1.200000, sum=11.200000
10 + 1.200000 = 11.200000
10_
```

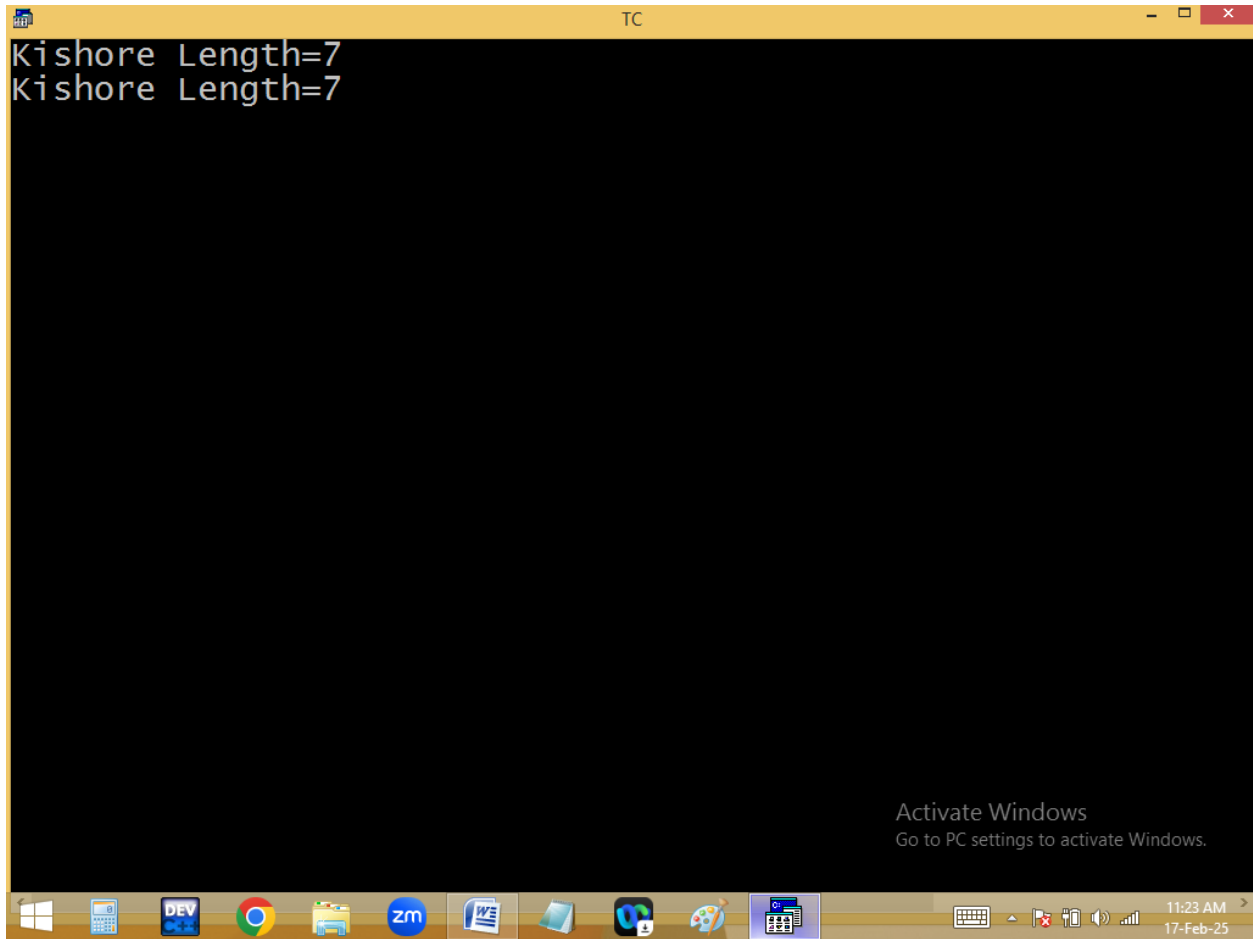
Activate Windows
Go to PC settings to activate Windows.

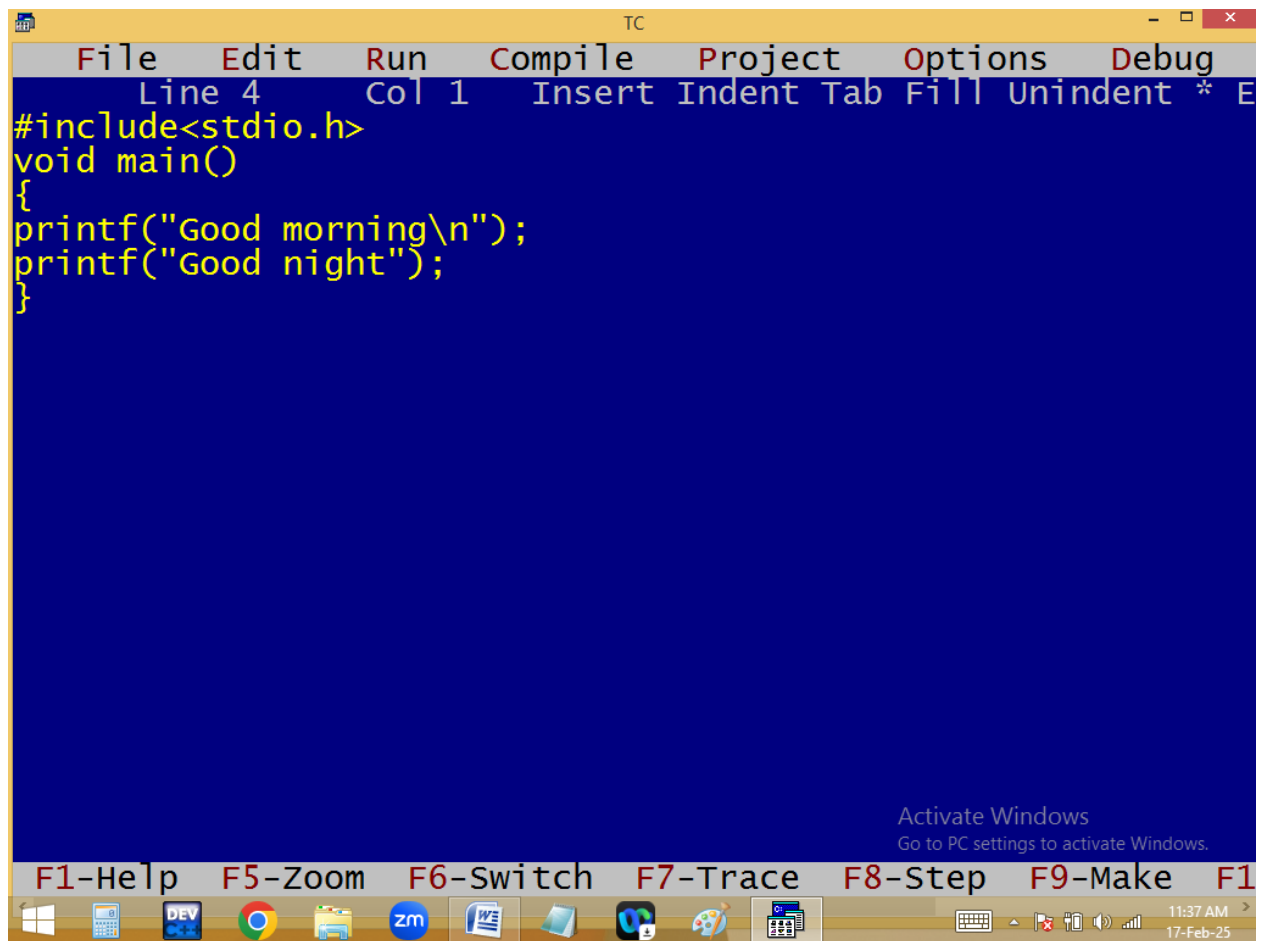
11:36 AM
17-Feb-25

Write a c program to find string length without using strlen() or a loop?

Eg: Kishore length 7







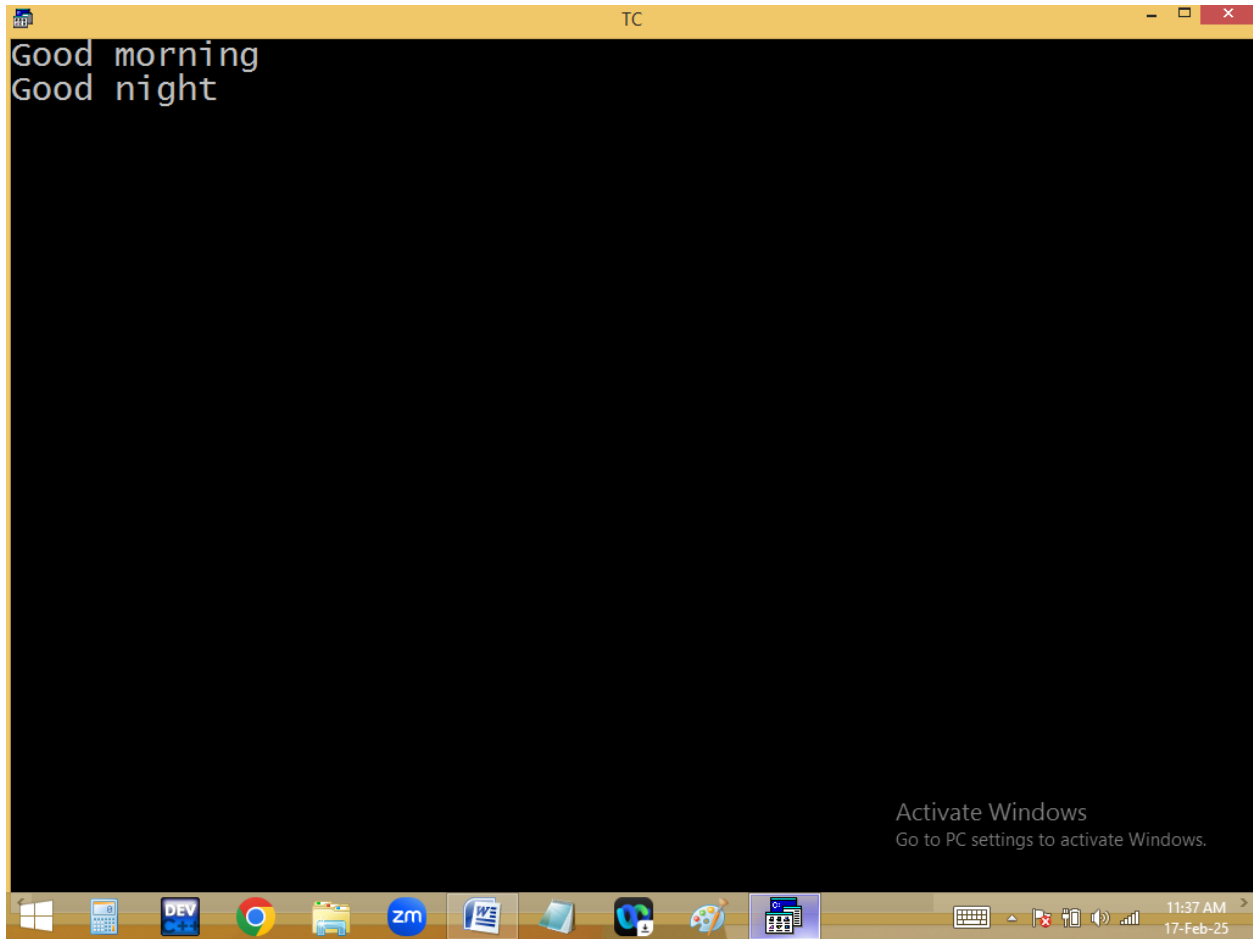
The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 4", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code in yellow text:

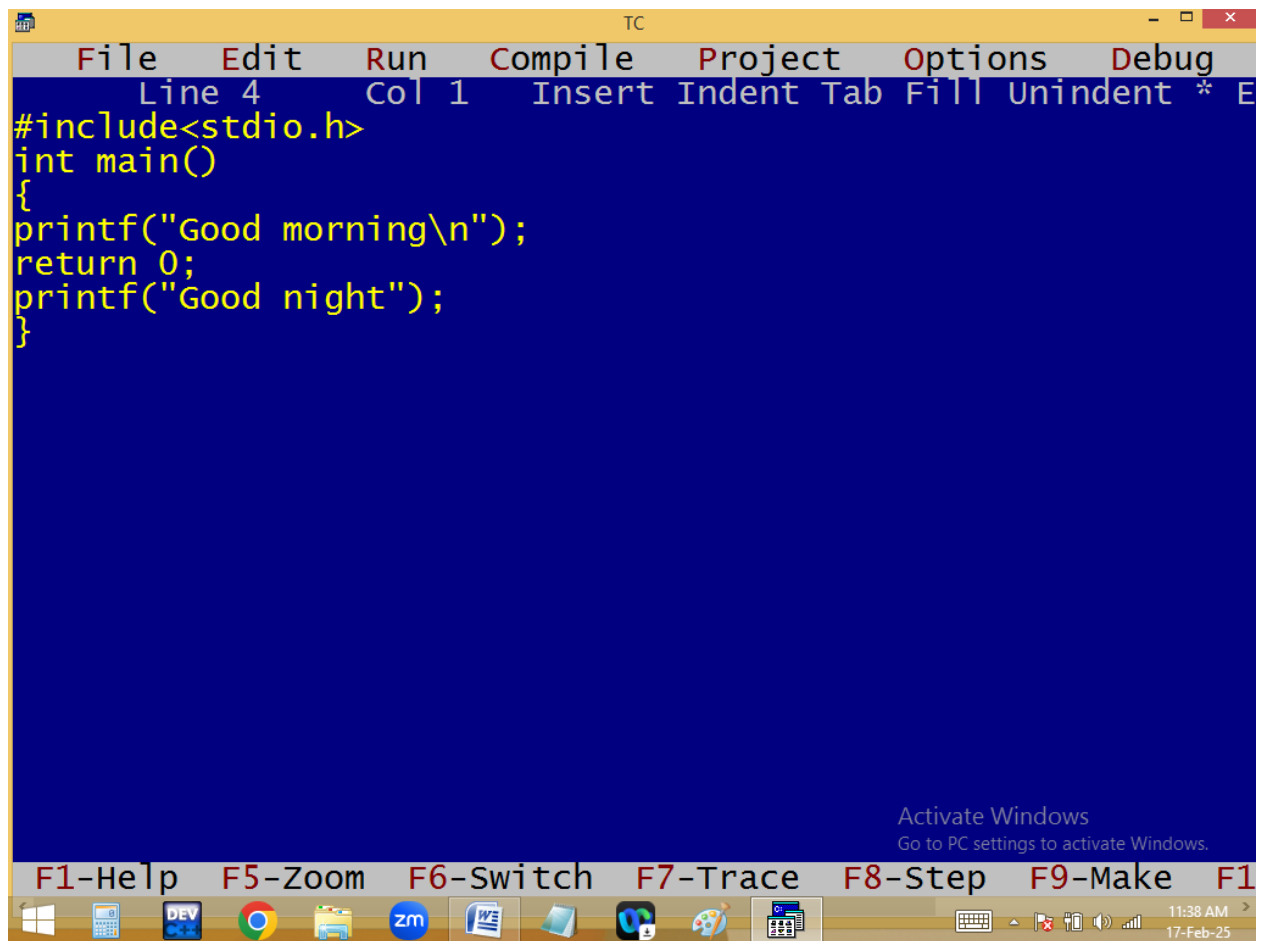
```
#include<stdio.h>
void main()
{
printf("Good morning\n");
printf("Good night");
}
```

At the bottom of the window, there is a taskbar with several icons: Windows Start button, Task View, File Explorer, Google Chrome, Dev C++, Zoho Mail, and a folder icon. To the right of these icons, there is a keyboard icon, a volume icon, and a network icon. The system clock shows "11:37 AM" and "17-Feb-25".

Below the menu bar, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit".

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, with the text "Go to PC settings to activate Windows."



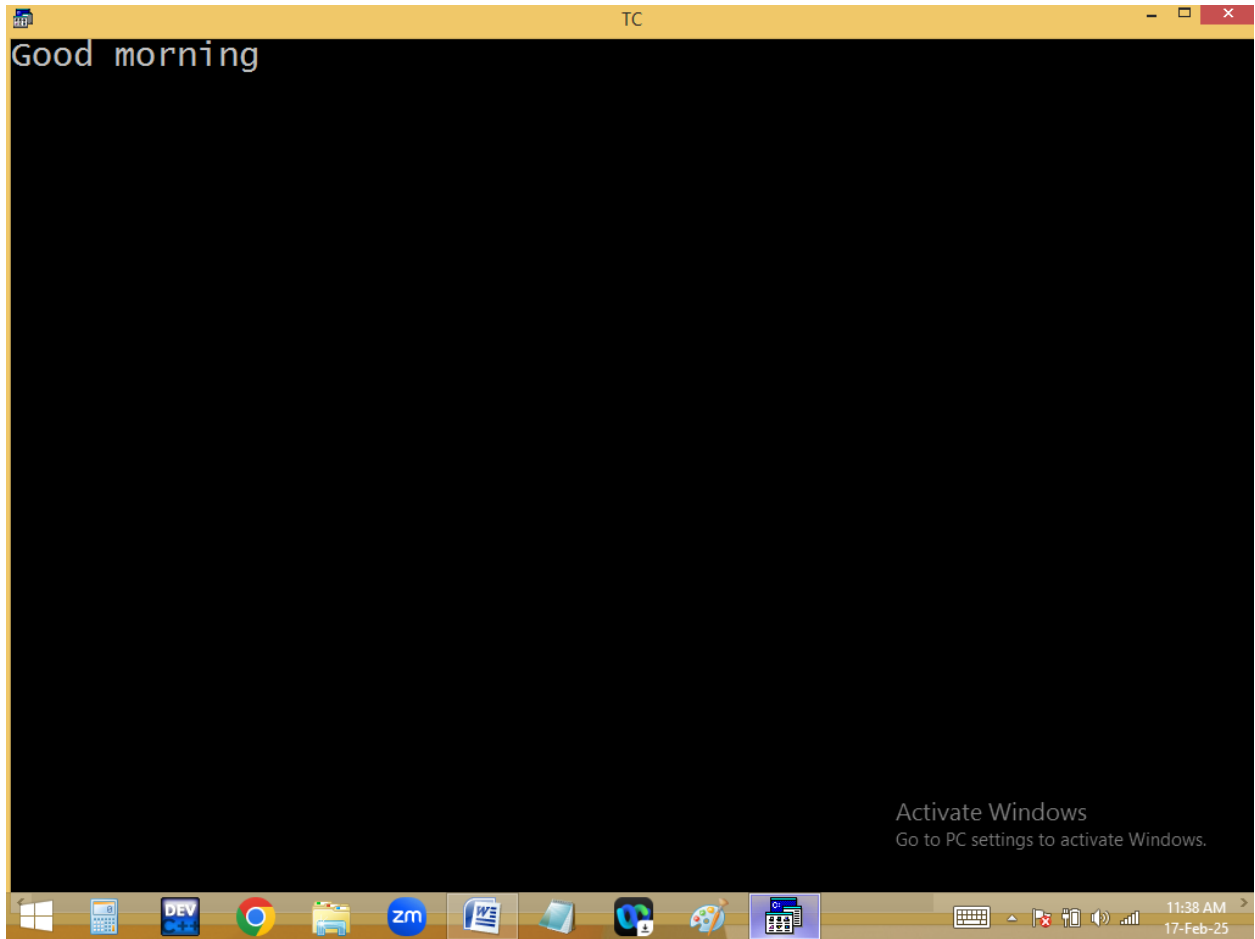


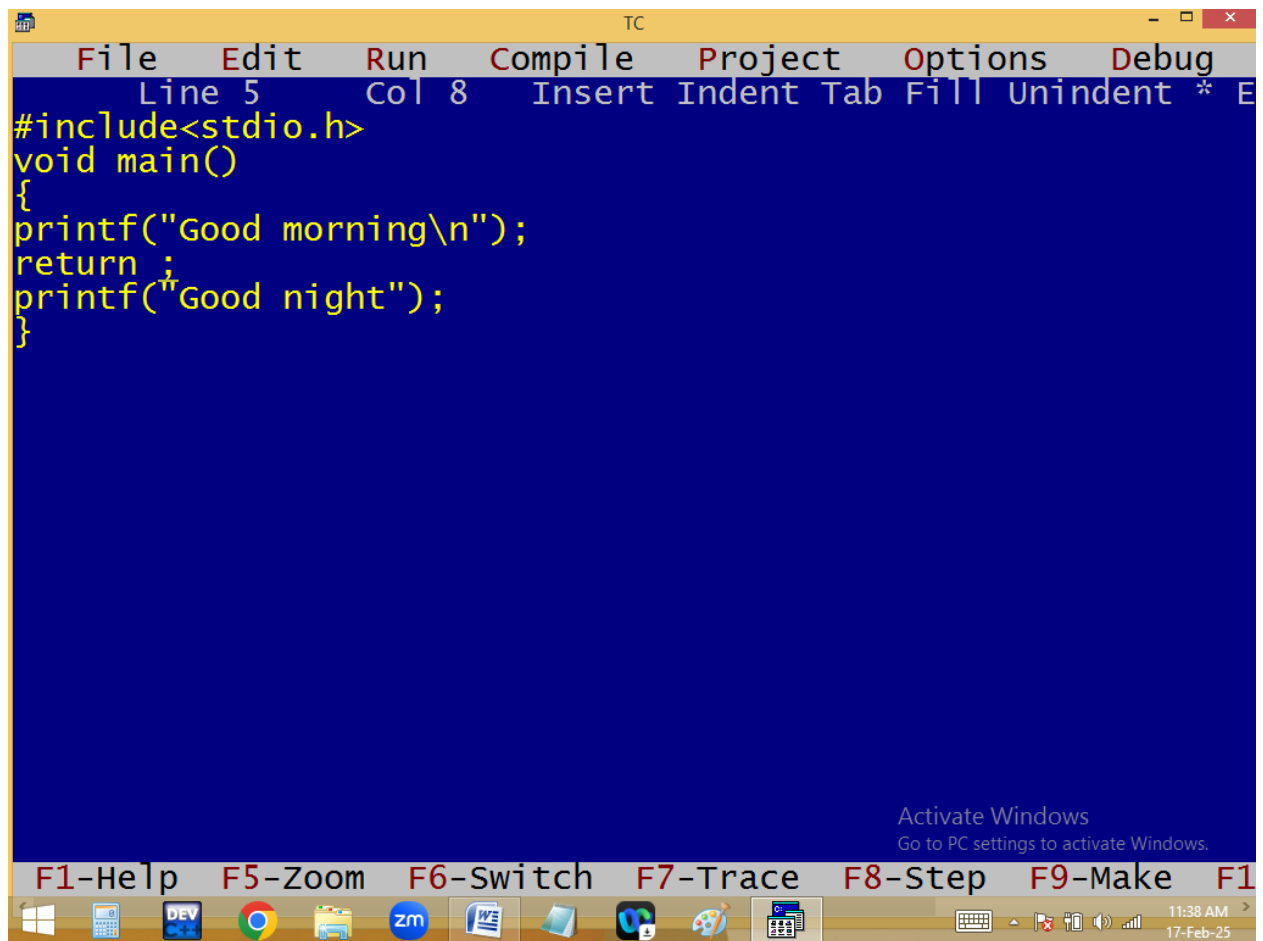
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 4", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
int main()
{
printf("Good morning\n");
return 0;
printf("Good night");
}
```

At the bottom of the window, there is a taskbar with several icons: Windows Start button, Task View, File Explorer, Google Chrome, Dev C++, Zoom, and a folder icon. The system tray on the right shows the time "11:38 AM" and the date "17-Feb-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

Below the IDE window, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit".





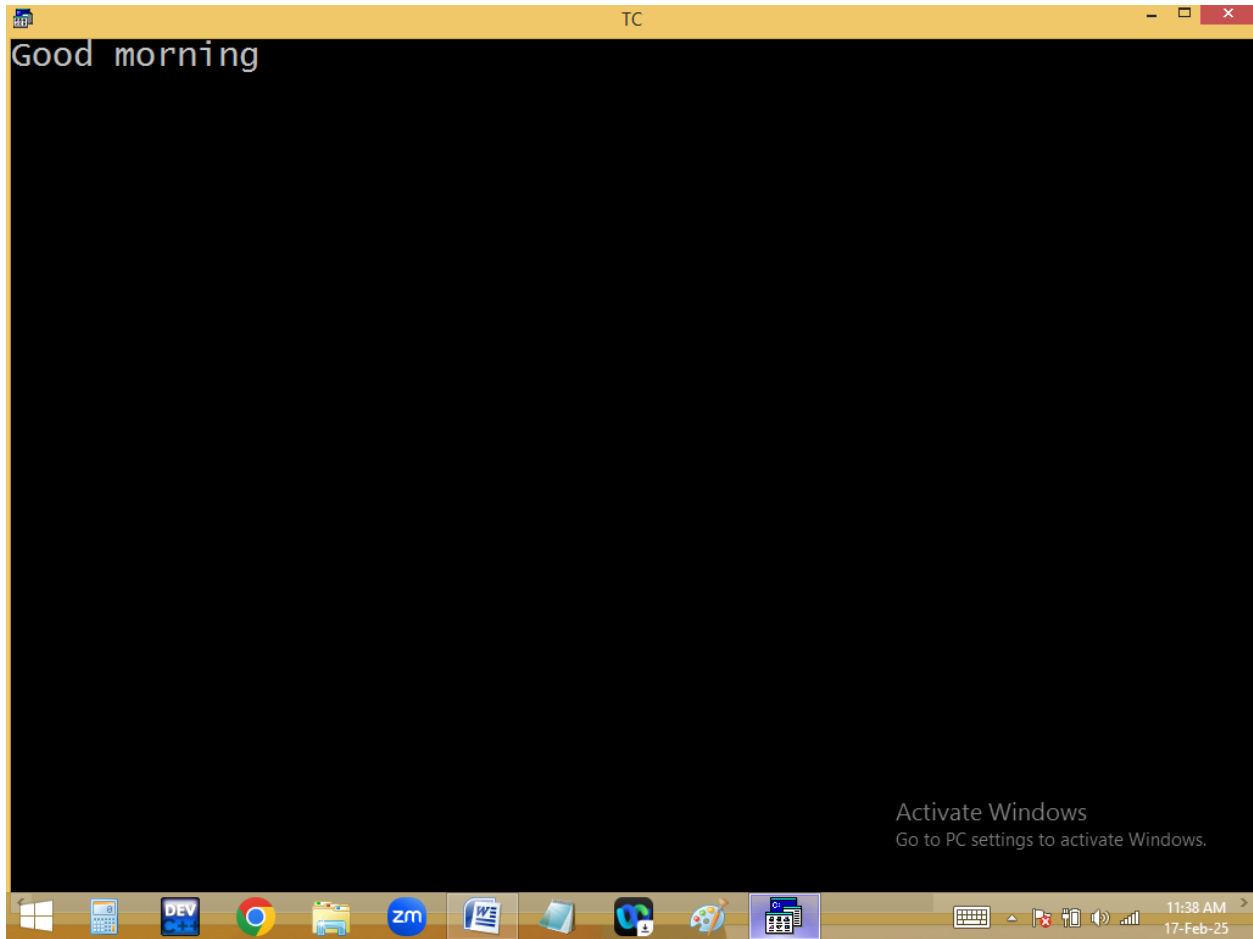
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 5", "Col 8", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

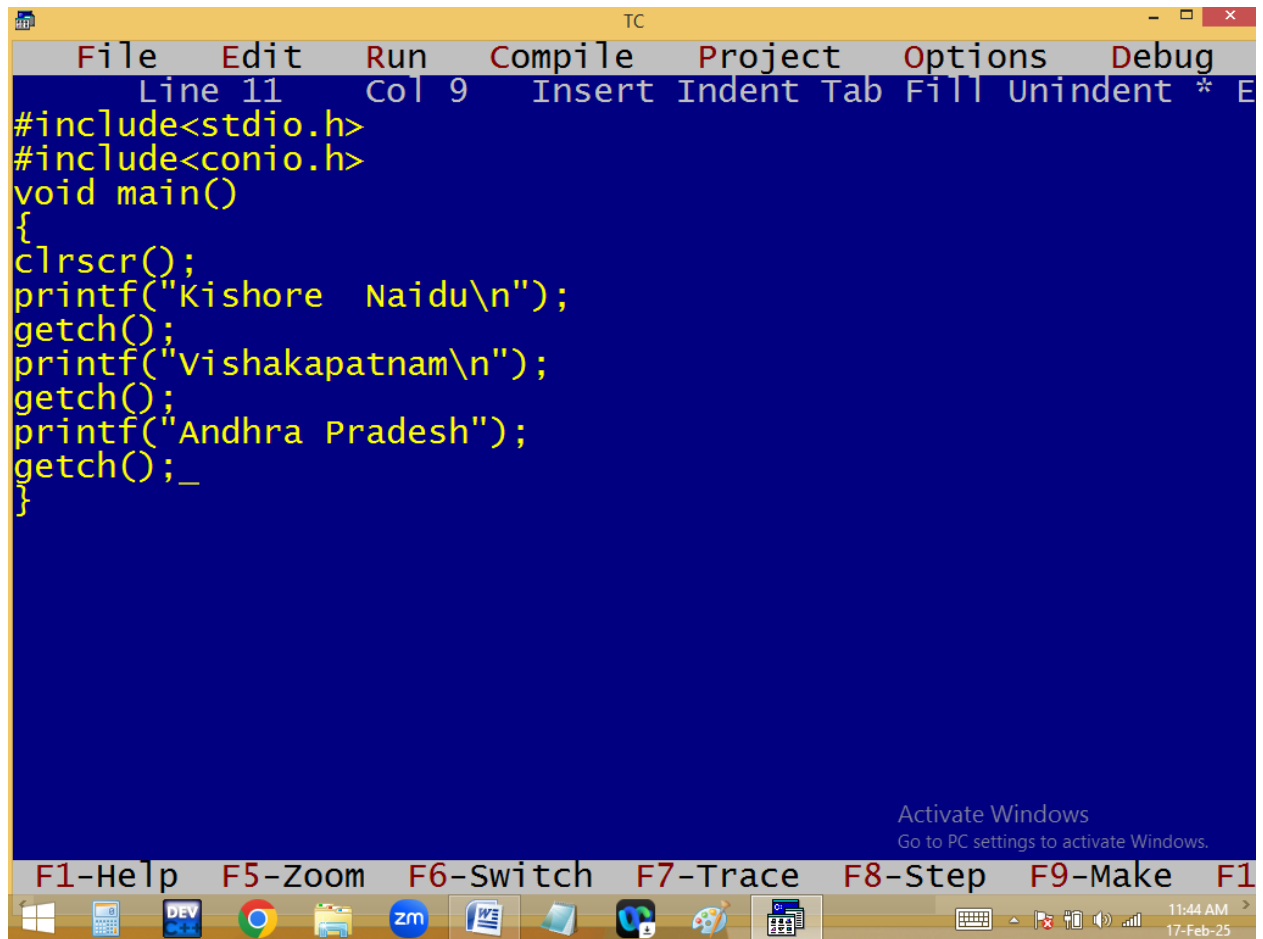
```
#include<stdio.h>
void main()
{
printf("Good morning\n");
return ;
printf("Good night");
}
```

At the bottom of the window, there is a taskbar with several icons: Windows Start button, Task View, File Explorer, Google Chrome, Dev C++, Zoom, and a folder icon. To the right of the taskbar, there is a system tray with icons for keyboard, volume, and network, along with the date and time "11:38 AM" and "17-Feb-25".

Below the code editor, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit".

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows."



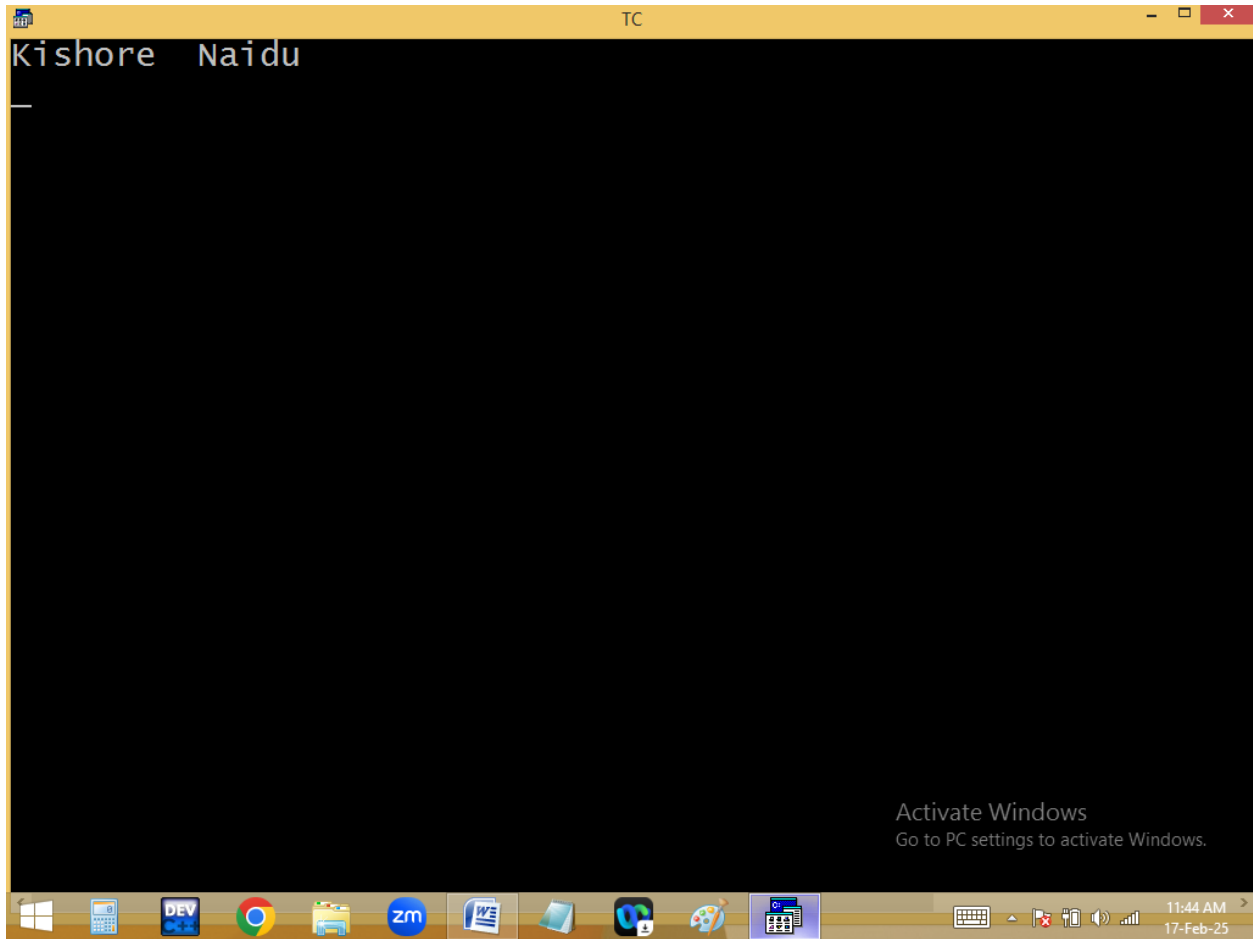


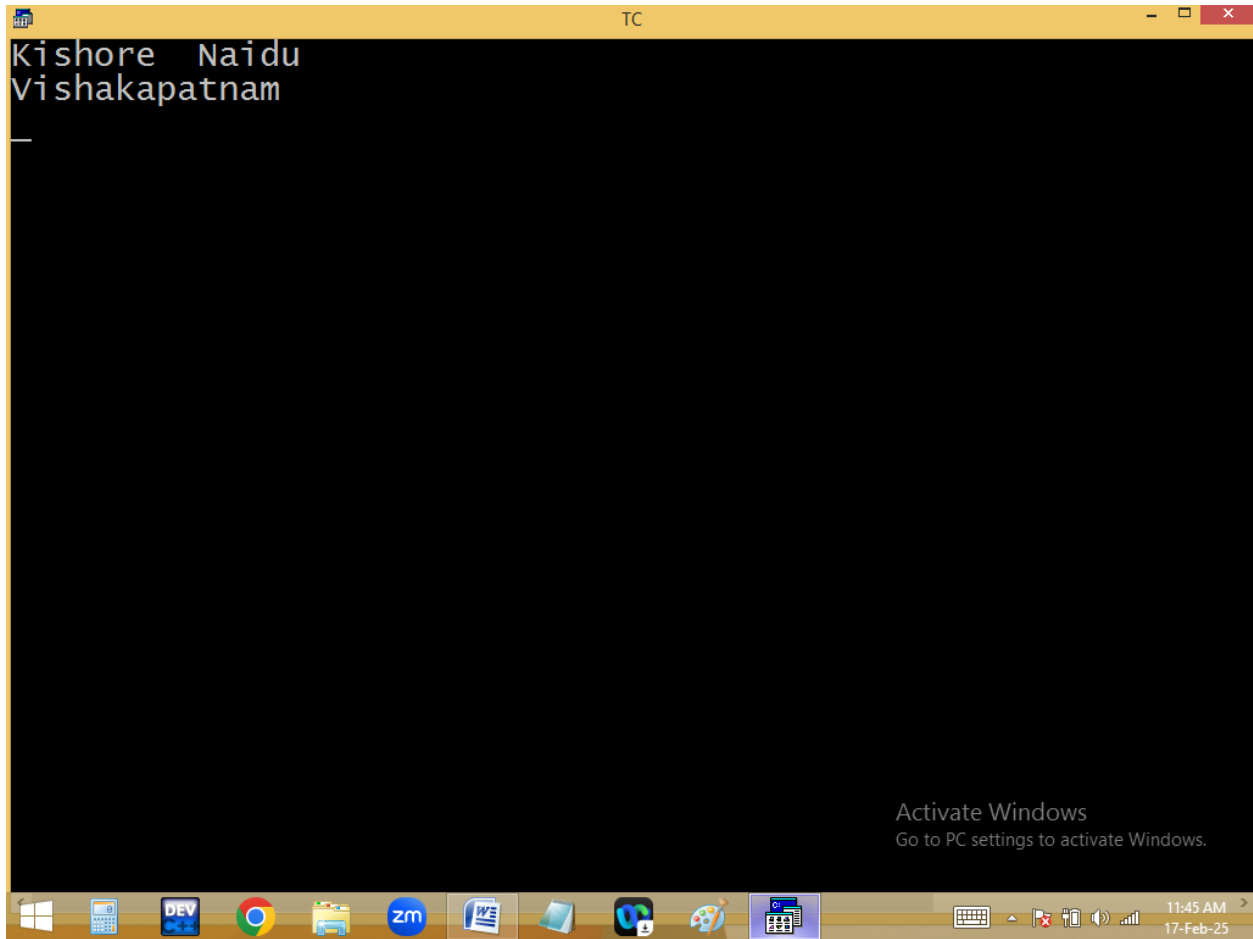
```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore Naidu\n");
getch();
printf("Vishakapatnam\n");
getch();
printf("Andhra Pradesh");
getch();_
}
```

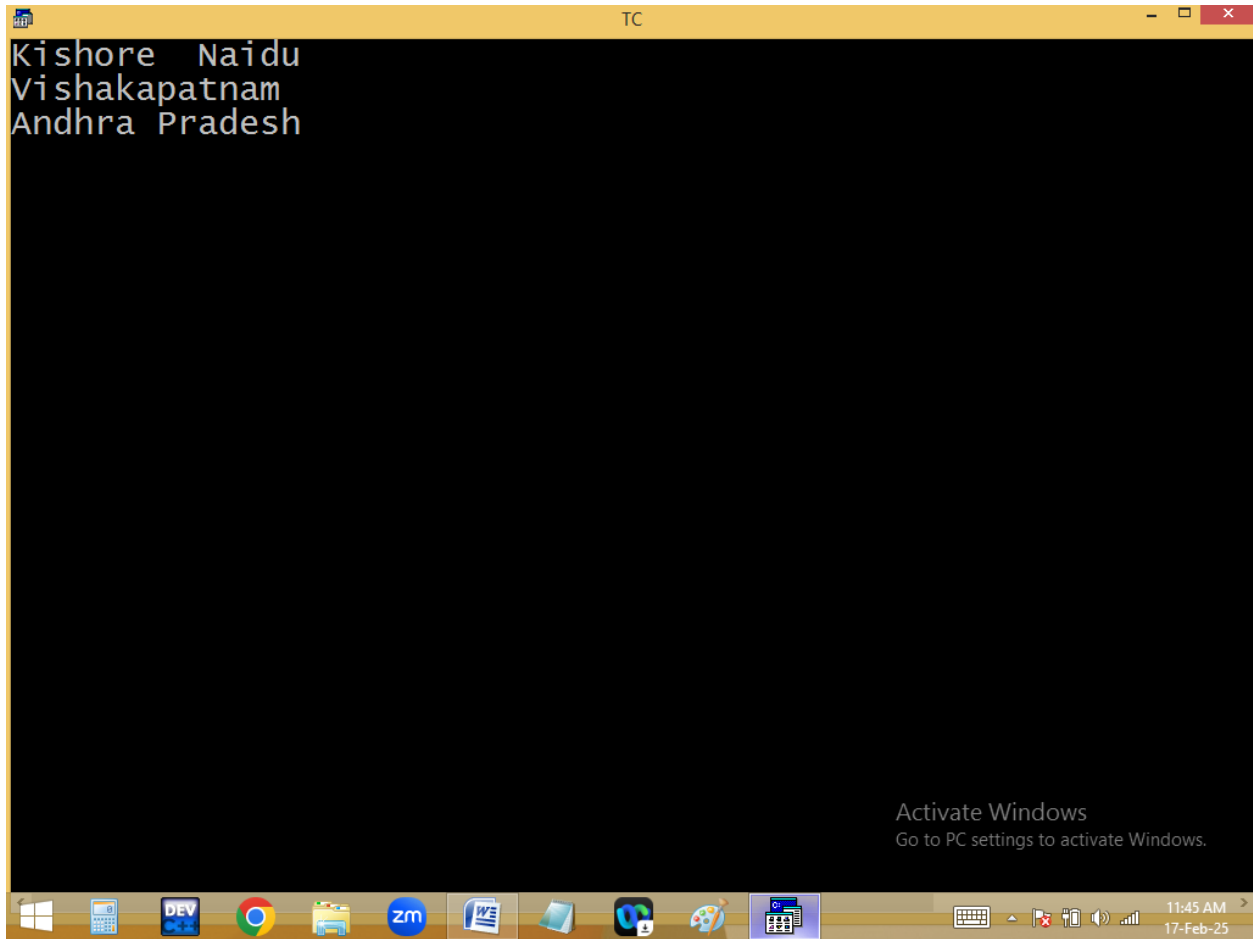
Activate Windows
Go to PC settings to activate Windows.

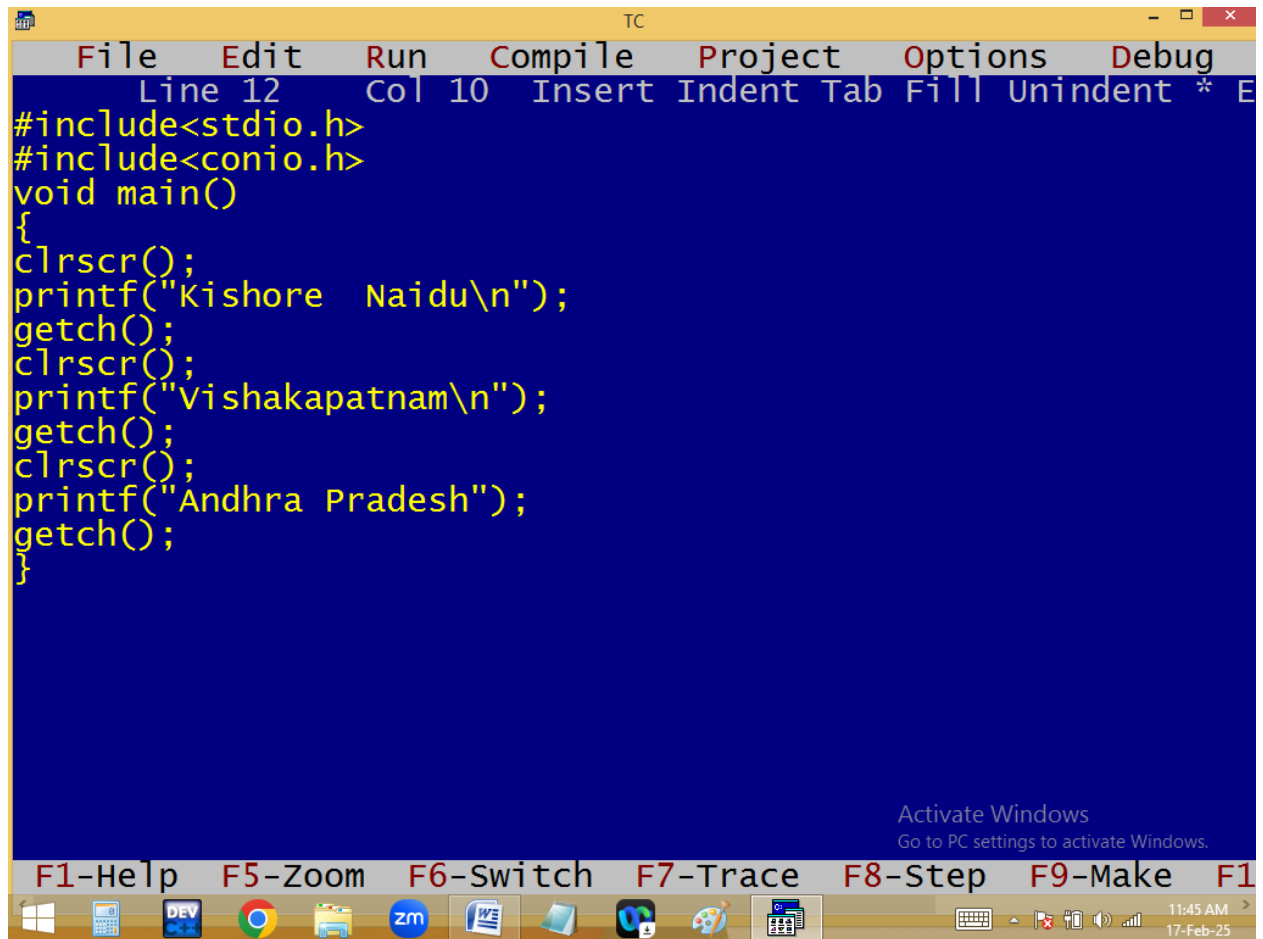
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

11:44 AM
17-Feb-25







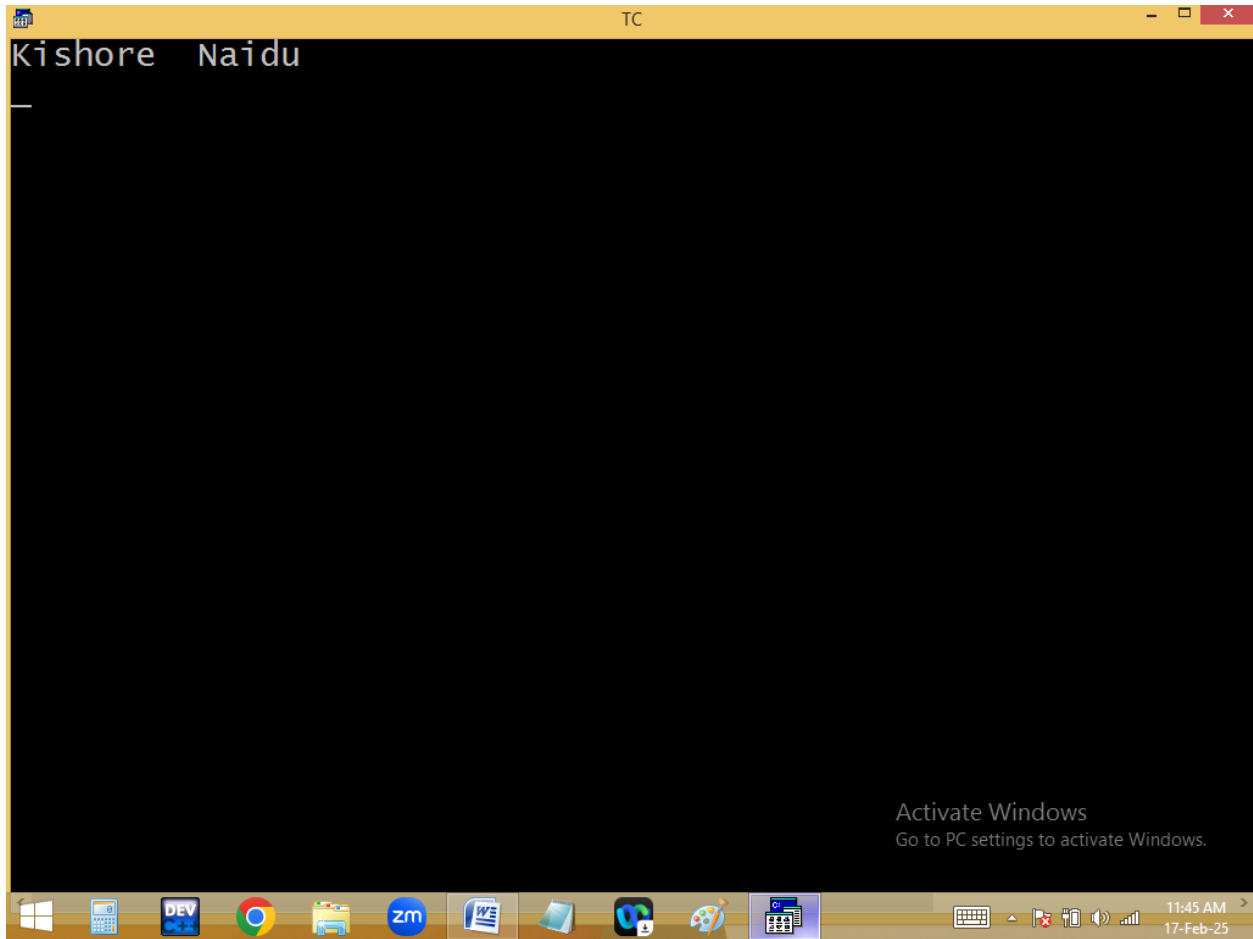


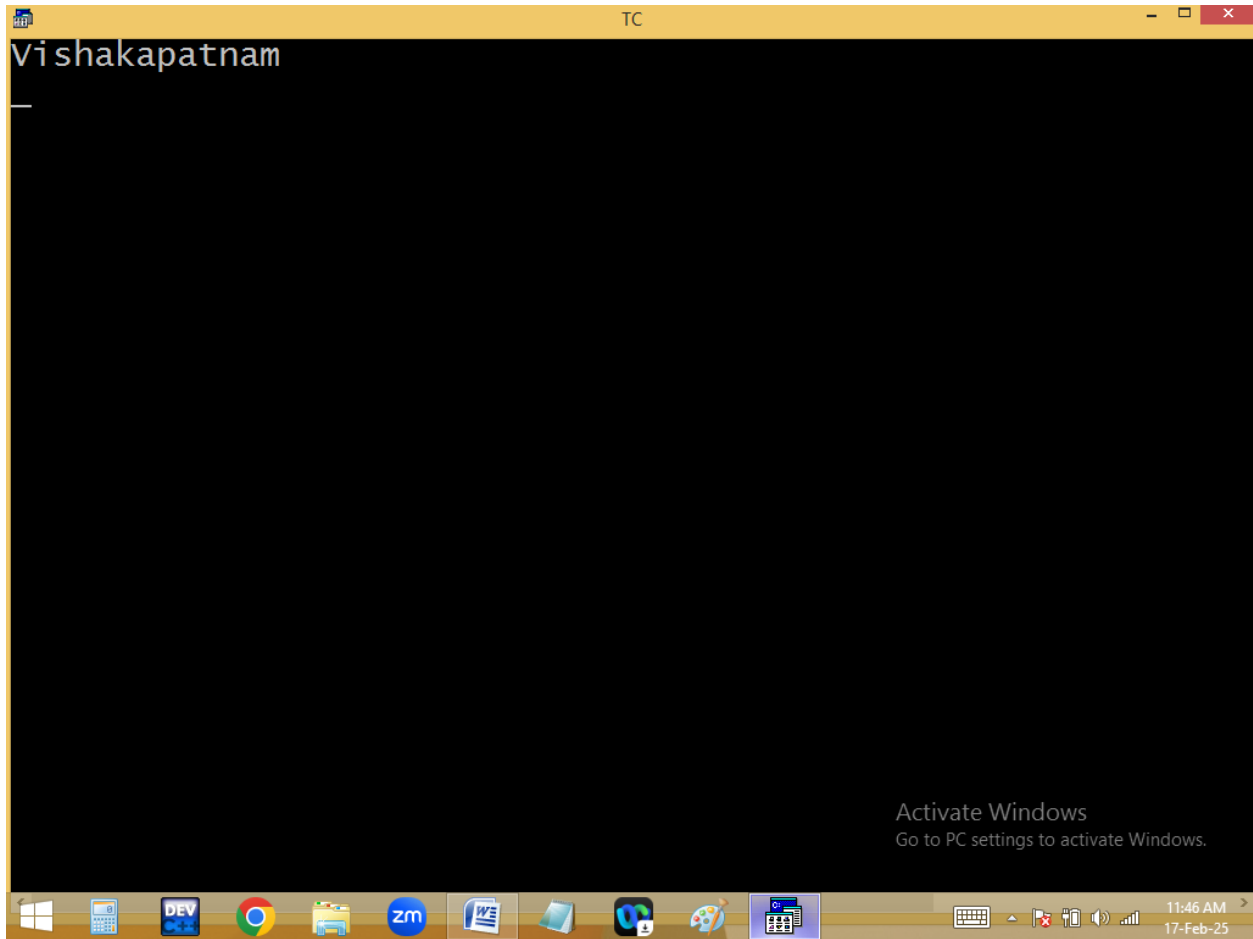
```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore Naidu\n");
getch();
clrscr();
printf("Vishakapatnam\n");
getch();
clrscr();
printf("Andhra Pradesh");
getch();
}
```

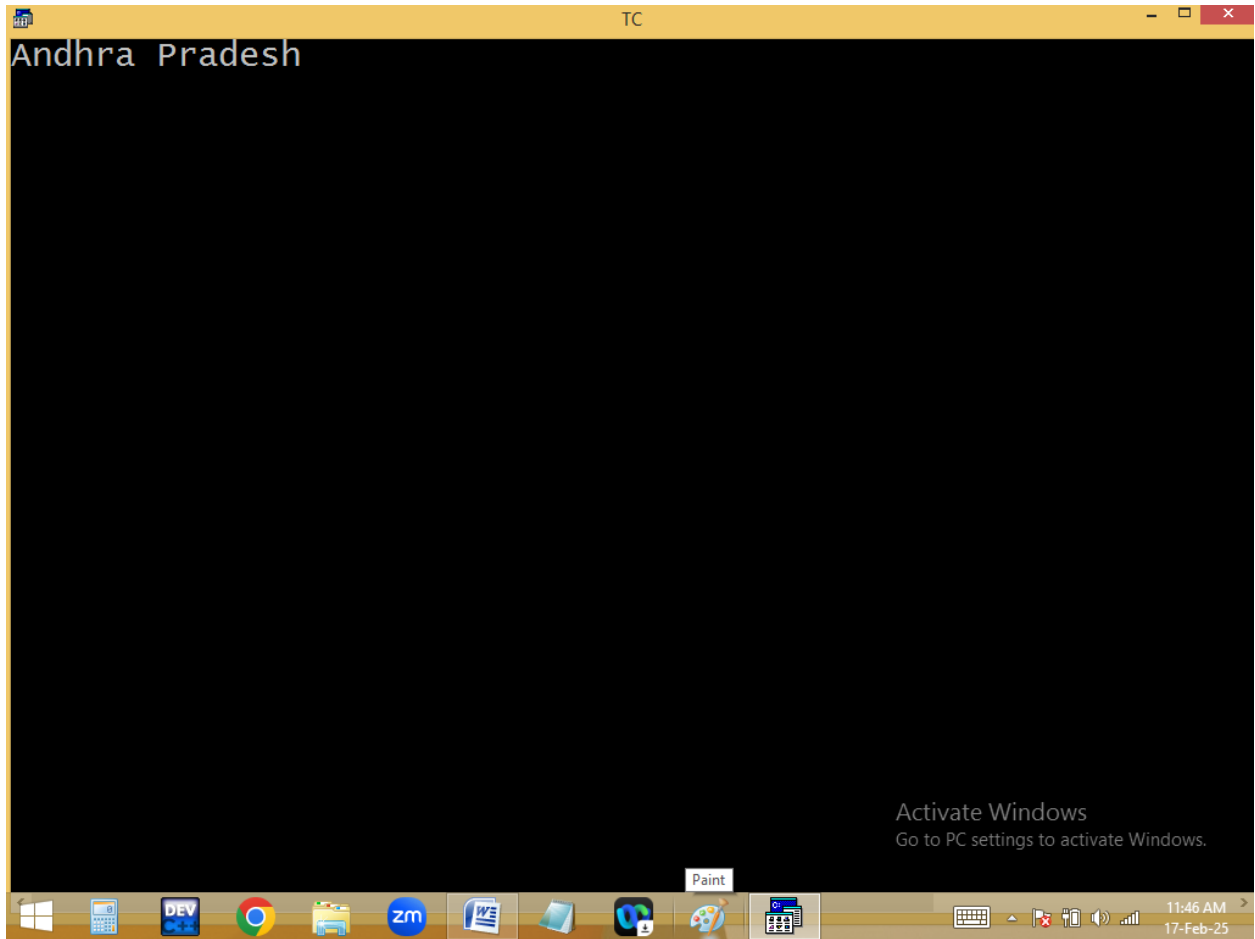
Activate Windows
Go to PC settings to activate Windows.

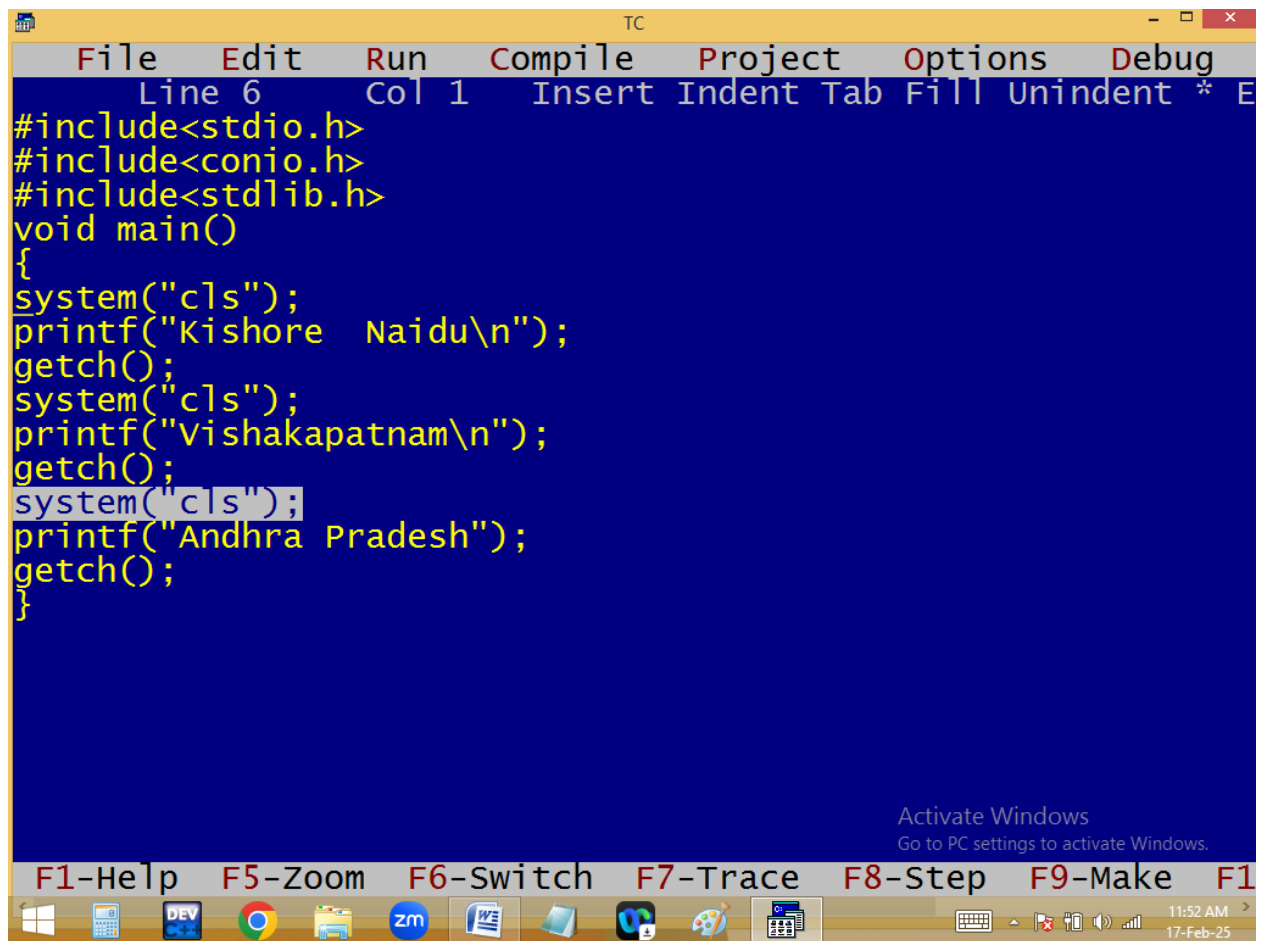
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

11:45 AM
17-Feb-25







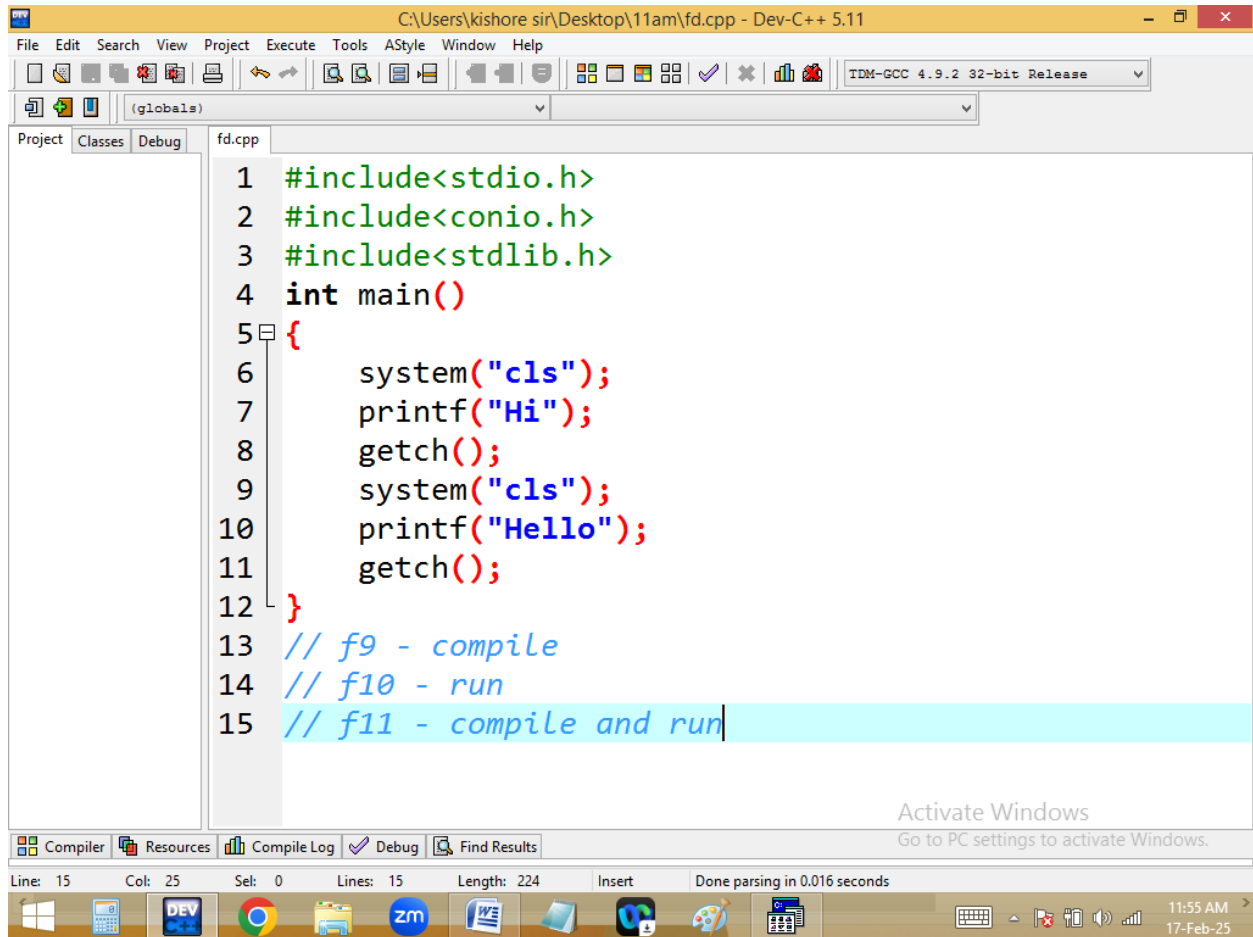


```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
system("cls");
printf("Kishore Naidu\n");
getch();
system("cls");
printf("Vishakapatnam\n");
getch();
system("cls");
printf("Andhra Pradesh");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

11:52 AM
17-Feb-25



```
1 #include<stdio.h>
2 #include<conio.h>
3 #include<stdlib.h>
4 int main()
5 {
6     system("cls");
7     printf("Hi");
8     getch();
9     system("cls");
10    printf("Hello");
11    getch();
12 }
13 // f9 - compile
14 // f10 - run
15 // f11 - compile and run
```

BACK SLASH / ESCAPE SEQUENCE CHARACTERS

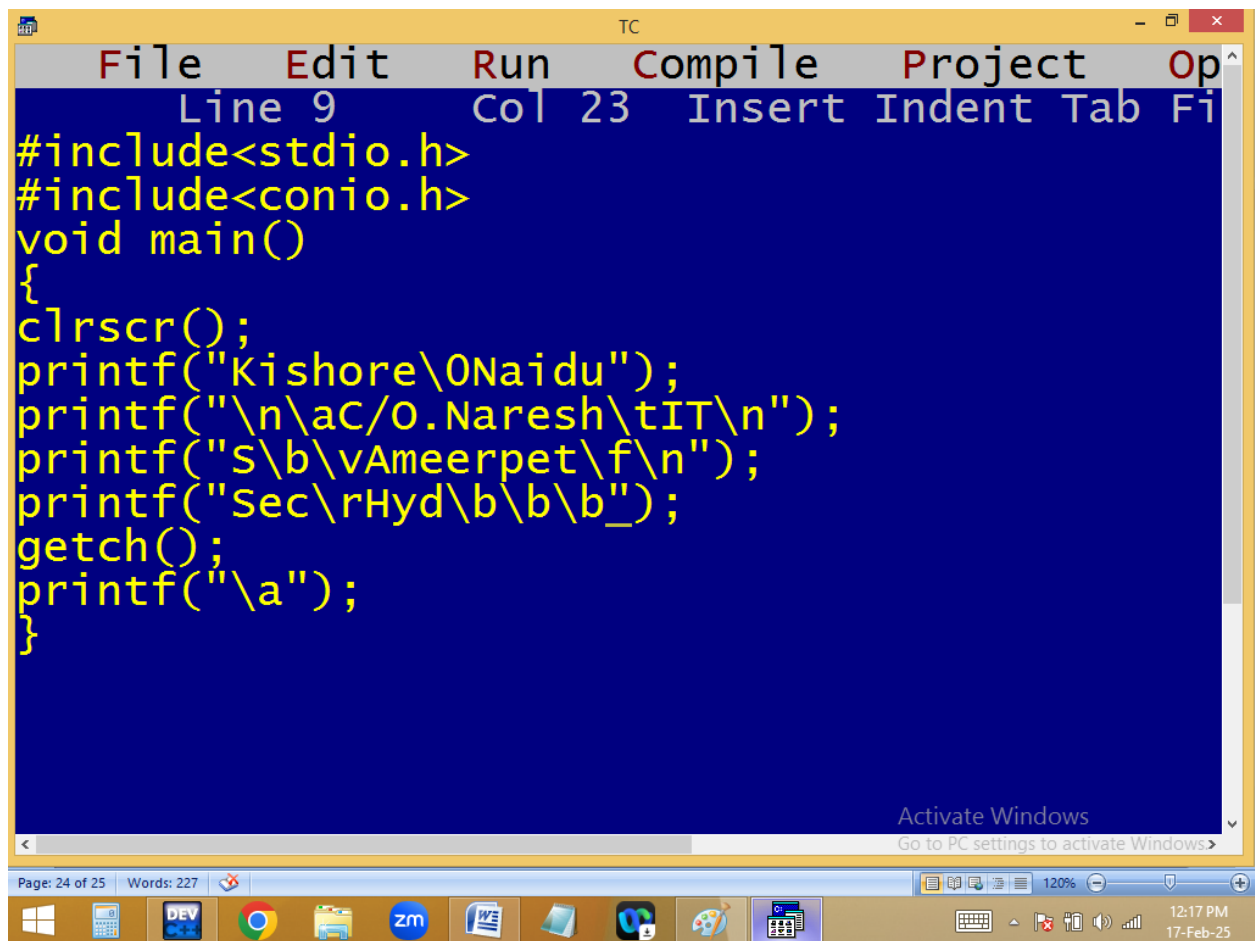
They started with back slash [\].

They used to format the outputs.

They participated in program execution but not displayed in output. Hence they are also called **escape sequence characters**.

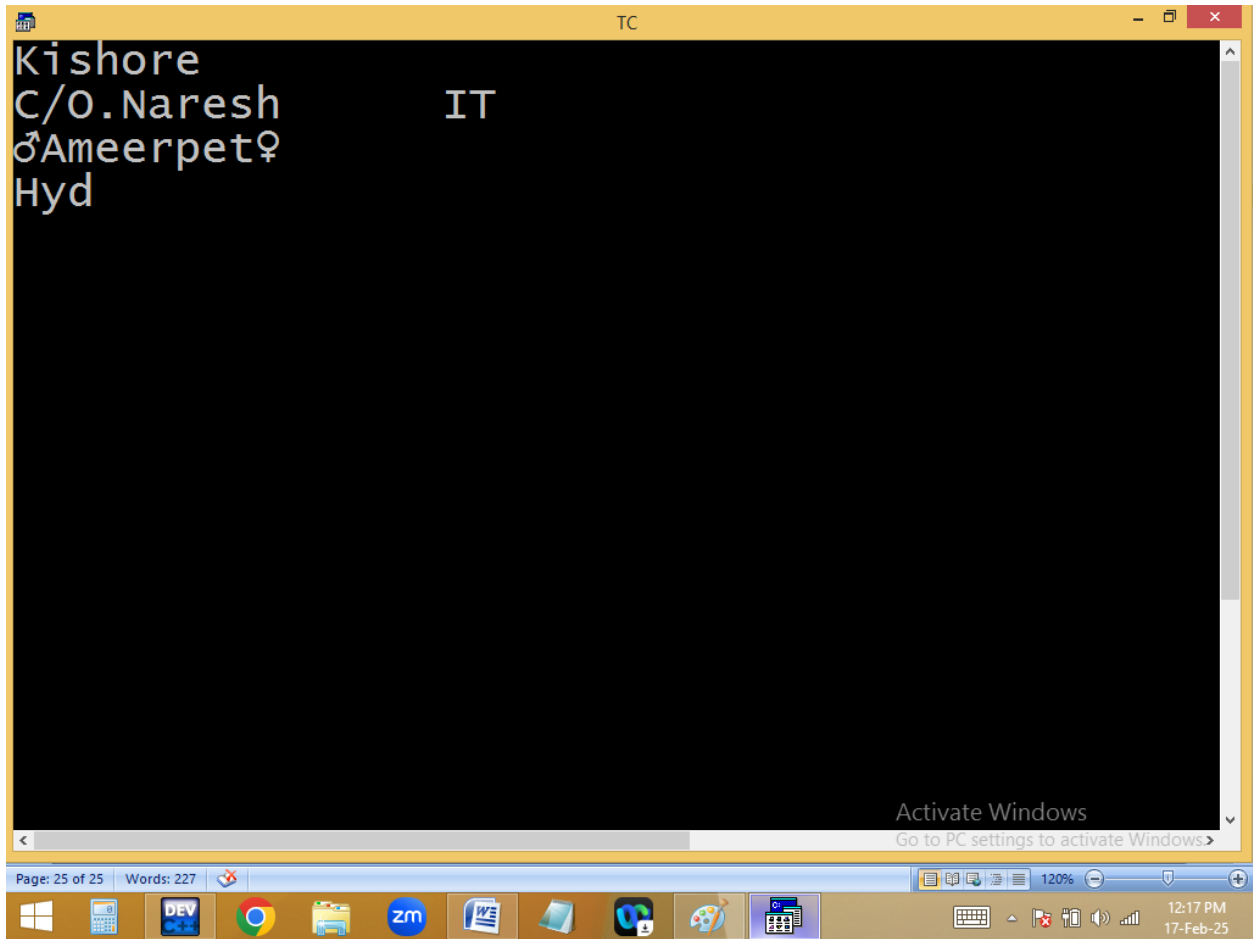
Each back slash character=**1 byte i.e. one character.**

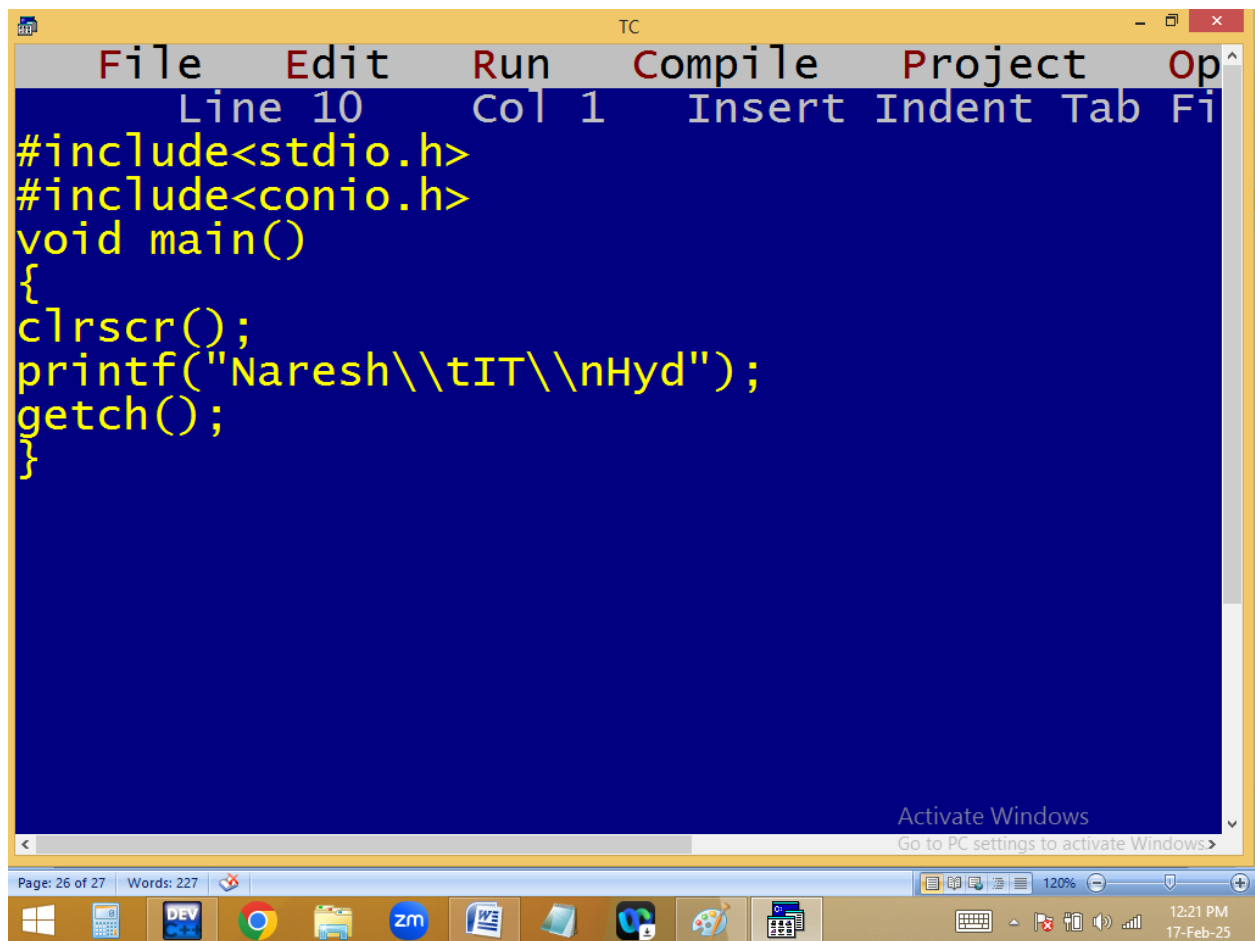
BACK SLASH CHARACTER	DESCRIPTION
\a	Alert [beep sound]
\b	Back space
\n	New line character
\t	Tab space
\r	Carriage return[beginning of line]
\f	Form feed ♀
\v	Vertical tab ♂
\0	Null char
\\	\ [invalid]
\k	k [invalid]



```
File Edit Run Compile Project Op
Line 9 Col 23 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore\0Naidu");
printf("\n\ac/O.Naresh\tIT\n");
printf("S\b\vAmeerpet\f\n");
printf("Sec\rHyd\b\b\b");
getch();
printf("\a");
}
```

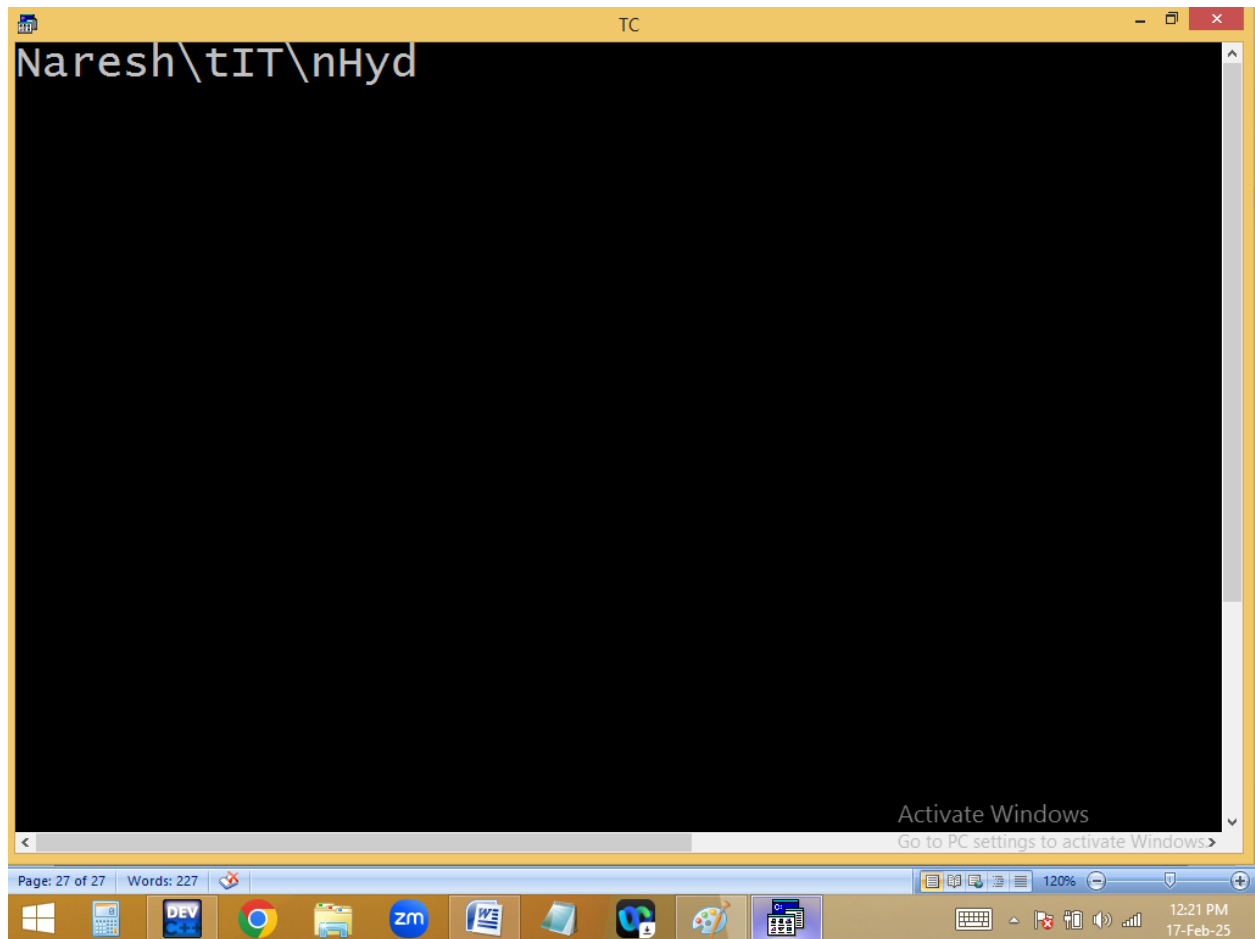
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```
File Edit Run Compile Project Op
Line 10 Col 1 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Naresh\\tIT\\nHyd");
getch();
}
```

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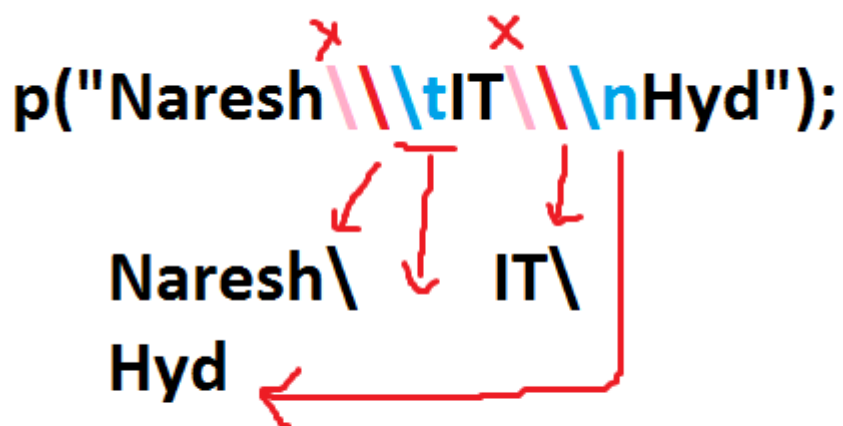


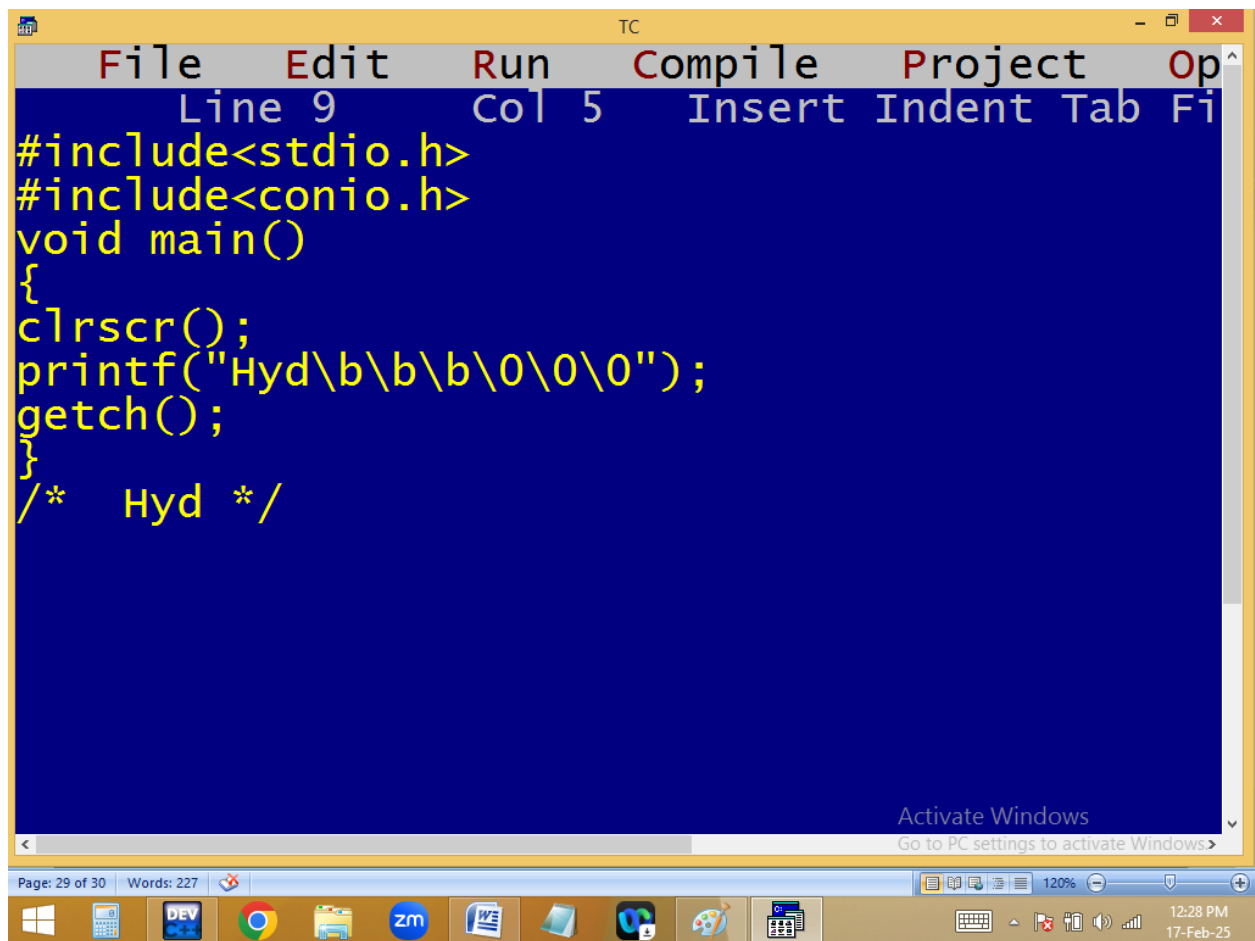
~~Naresh\tIT\nHyd~~

Naresh\tIT\nHyd

```
TC
File Edit Run Compile Project Op
Line 10 Col 10 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Naresh\\tIT\\nHyd");
getch();
}
/* Naresh\      IT\
Hyd */_

Activate Windows
Go to PC settings to activate Windows>
```



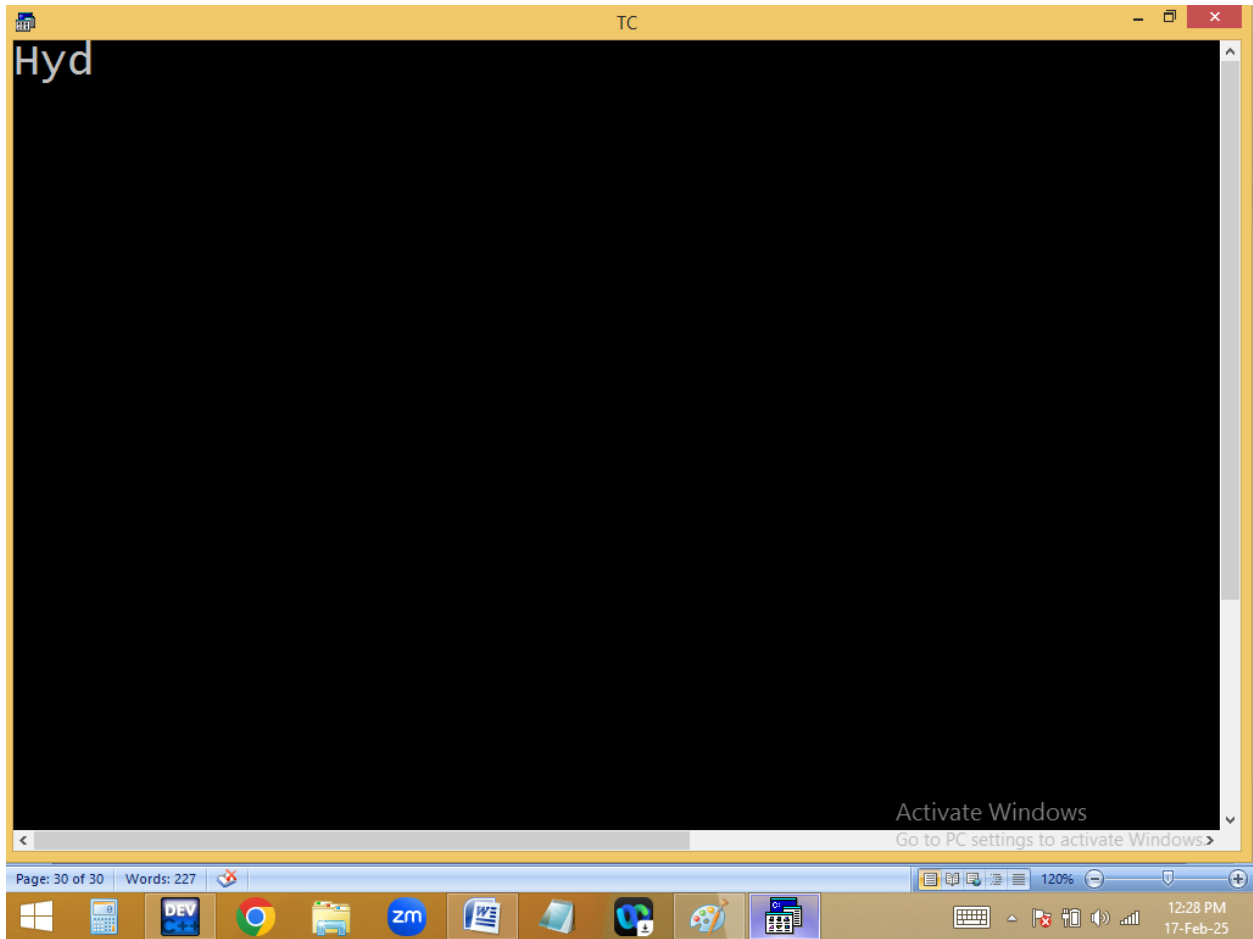


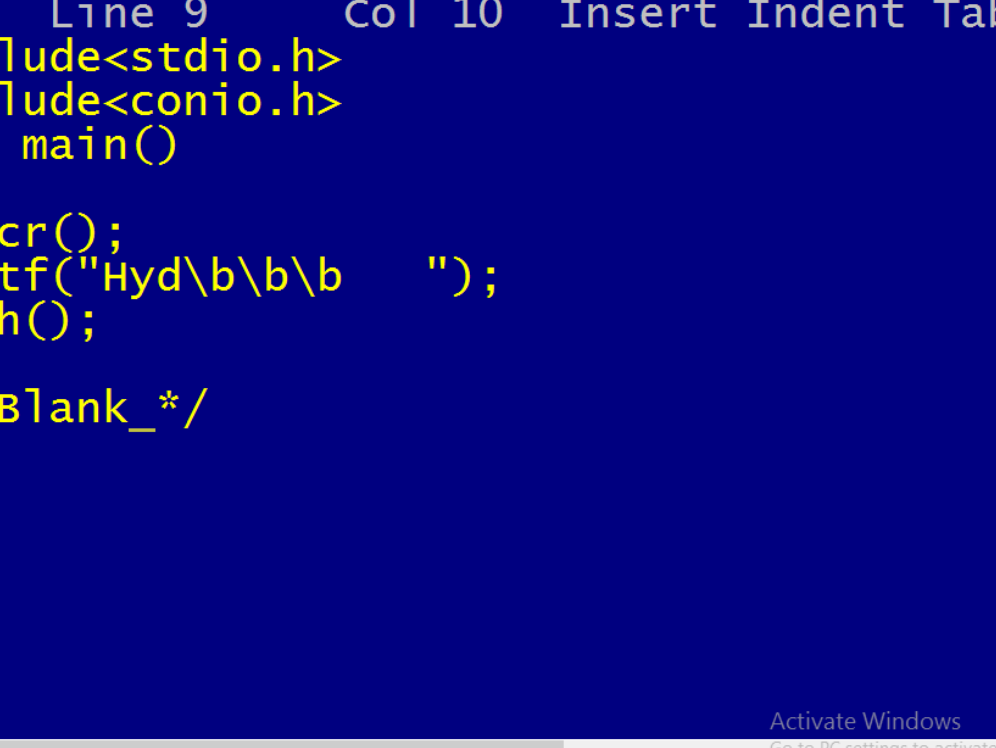
```
File Edit Run Compile Project Op
Line 9 Col 5 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Hyd\b\b\b\0\0\0");
getch();
}
/* Hyd */

Activate Windows
Go to PC settings to activate Windows>
```

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17-Feb-25

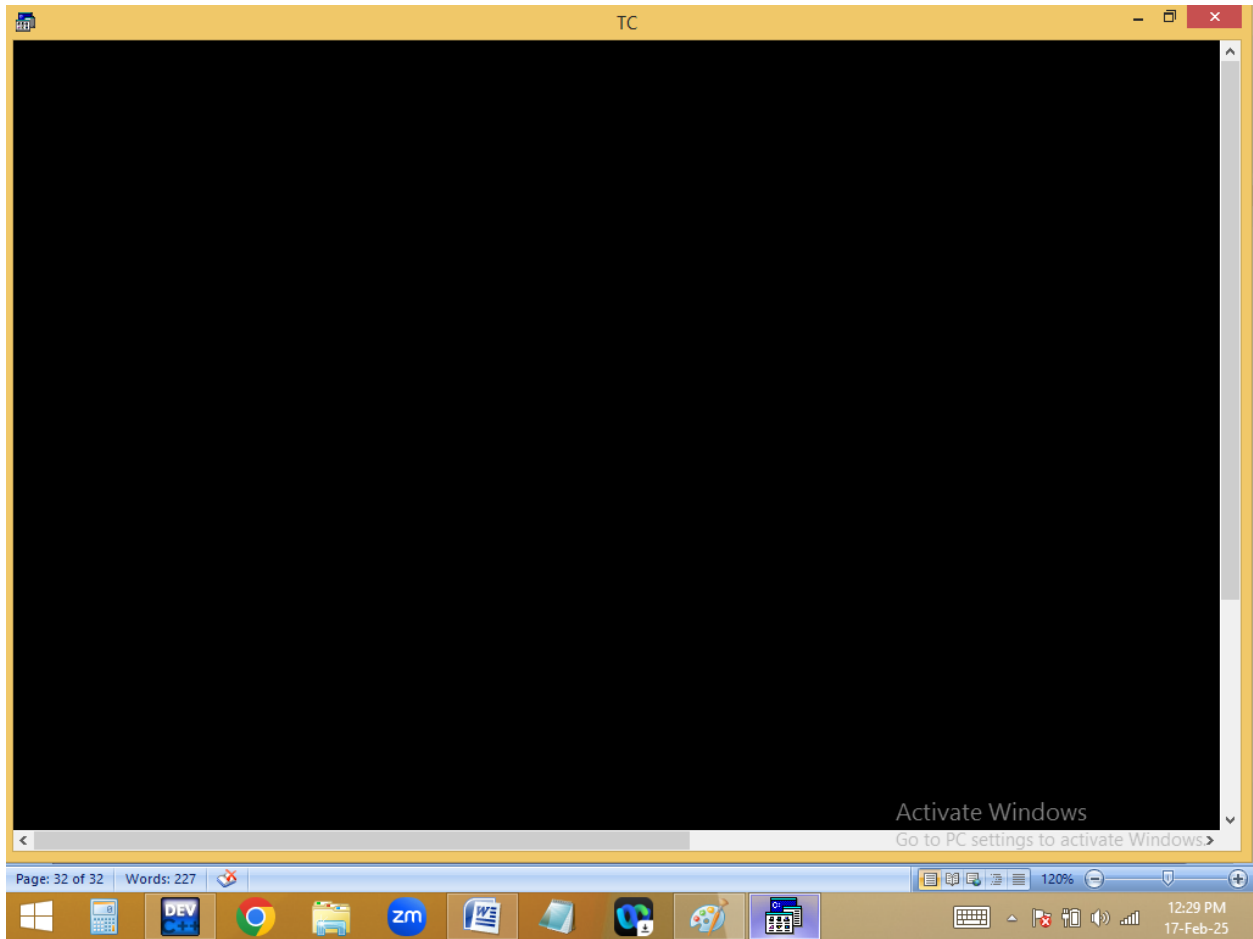


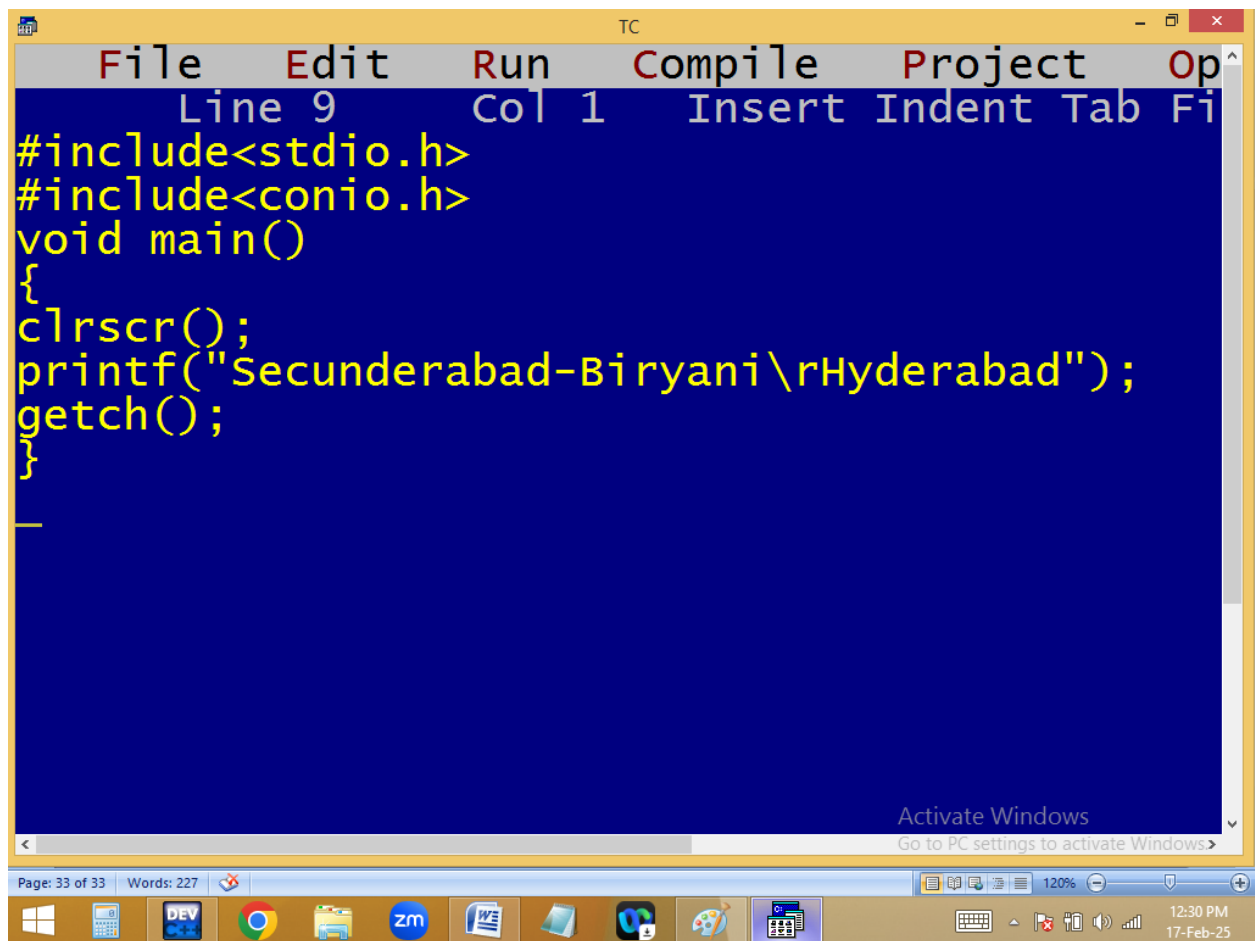


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar at the top includes File, Edit, Run, Compile, Project, and Op. The status bar at the top indicates Line 9, Col 10, and Insert mode. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Hyd\\b\\b\\b  ");
getch();
}
/* Blank_*/
```

The code is written in yellow text. The output of the program, "Hyd\\b\\b\\b", is visible on the screen. The Windows taskbar at the bottom shows the time as 12:29 PM on 17-Feb-25. The Windows Start button is visible on the left side of the taskbar.

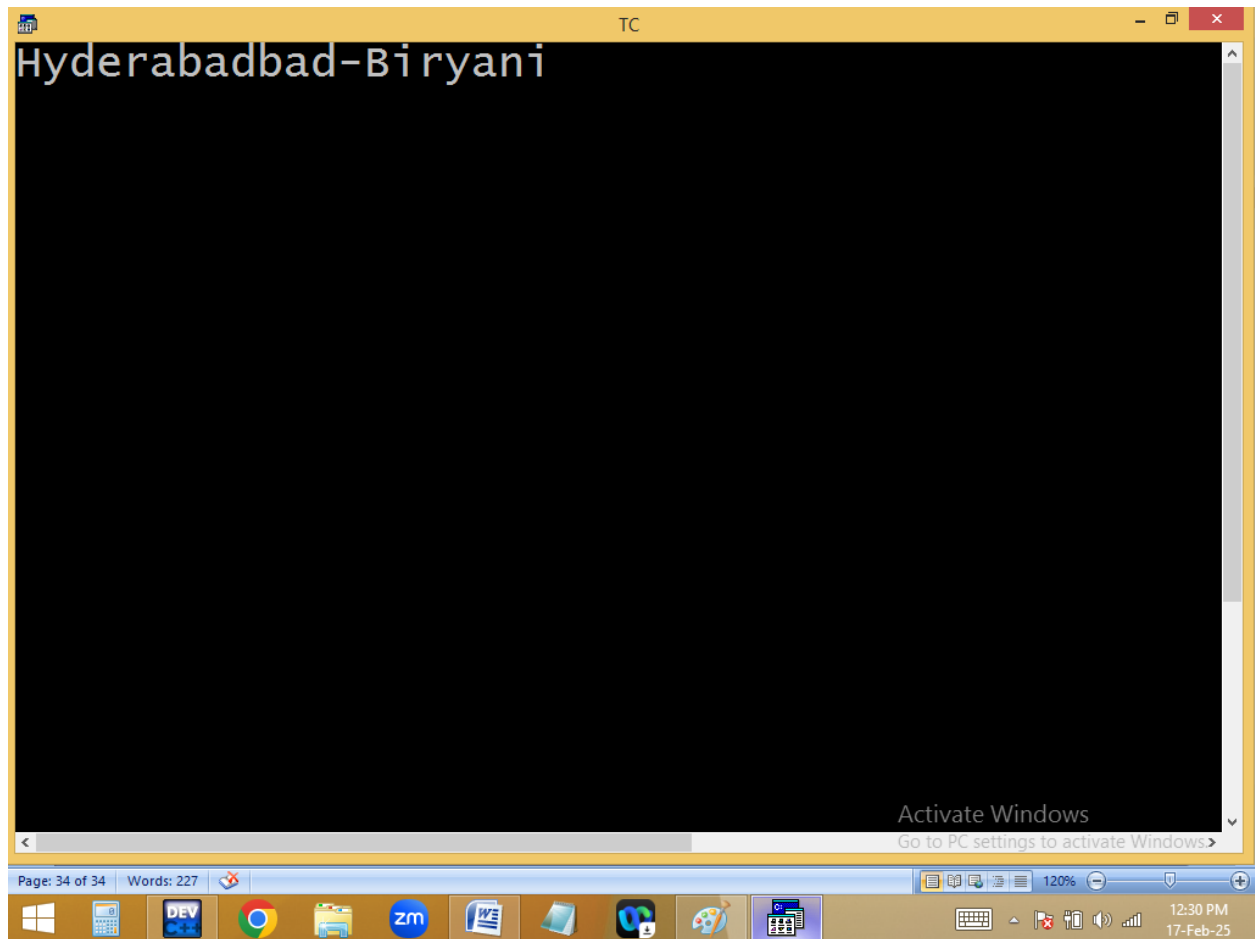




```
File Edit Run Compile Project Op
Line 9 Col 1 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Secunderabad-Biryani\rHyderabad");
getch();
}
```

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12:30 PM
17-Feb-25

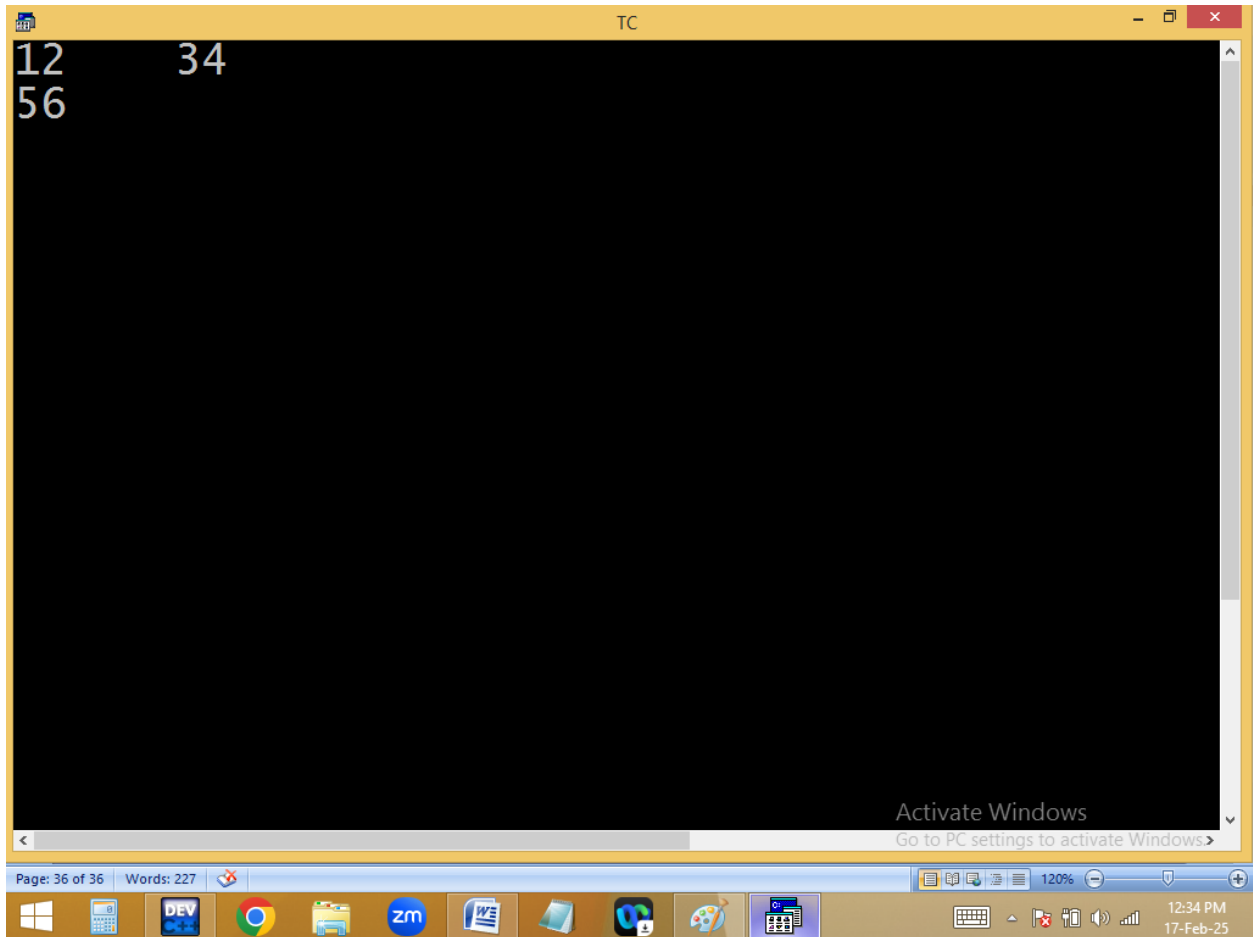


```
p("Secunderabad-Biryani  
\\rHyderabad");
```

```
File Edit Run Compile Project Op
Line 9 Col 1 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("12%c34%c56",9,10);
getch();
}
```

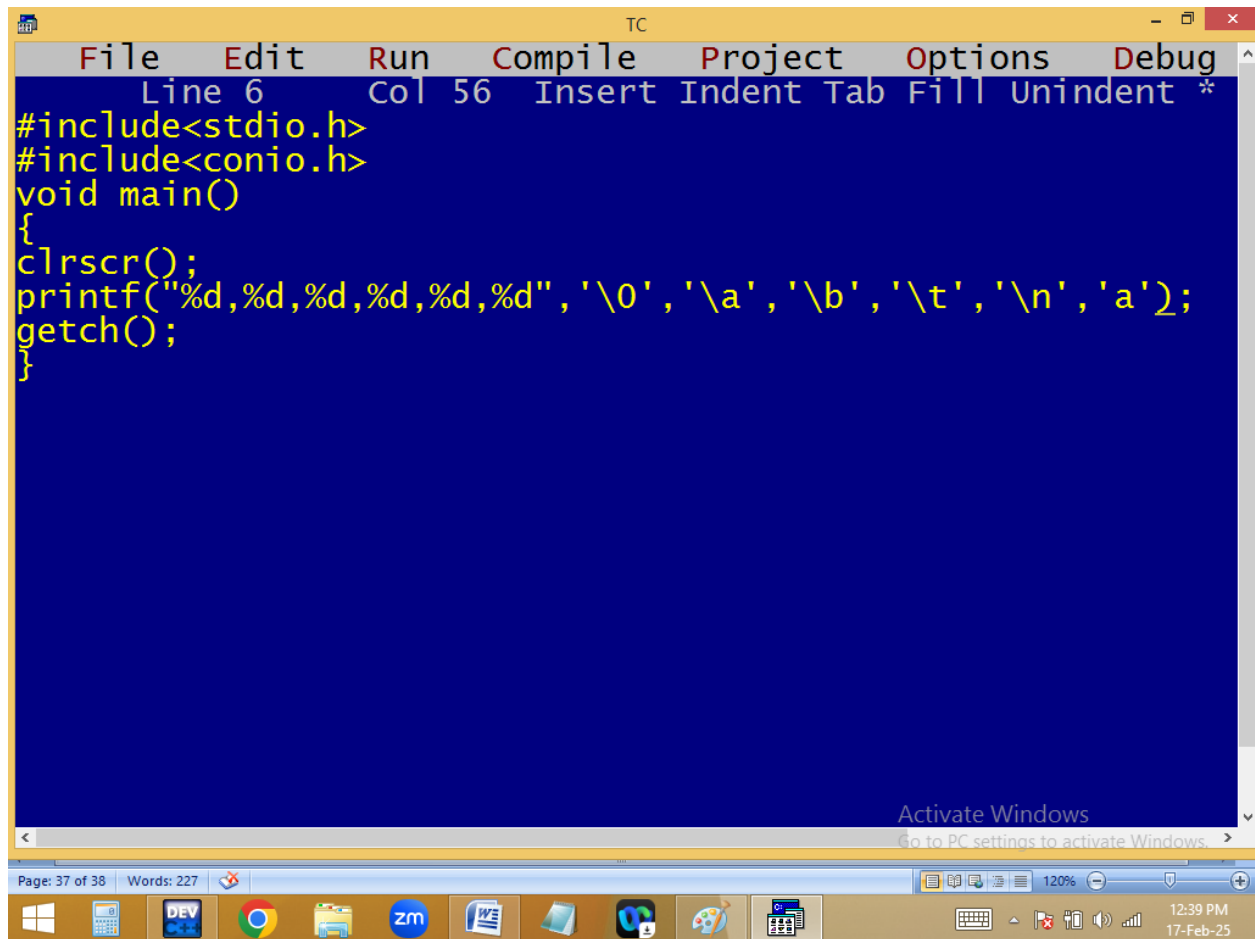
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17-Feb-25



p("12%c34%c56", 9, 10);

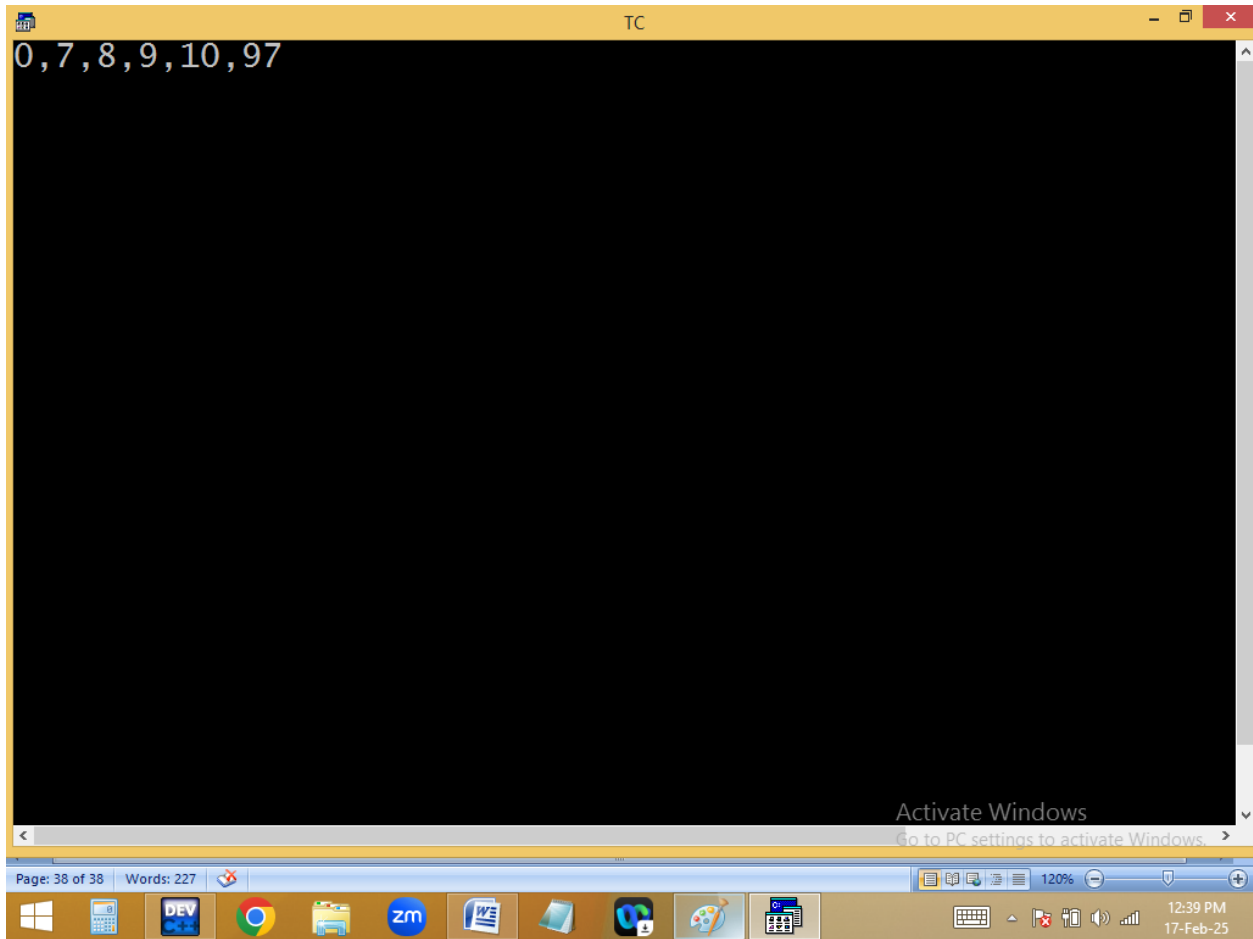
Hand-drawn annotations on the code snippet. An orange arrow points from the '9' to the '34' in the format string. A blue arrow points from the '10' to the '12' in the format string. There are also small orange and blue marks above the '9' and '10' respectively.

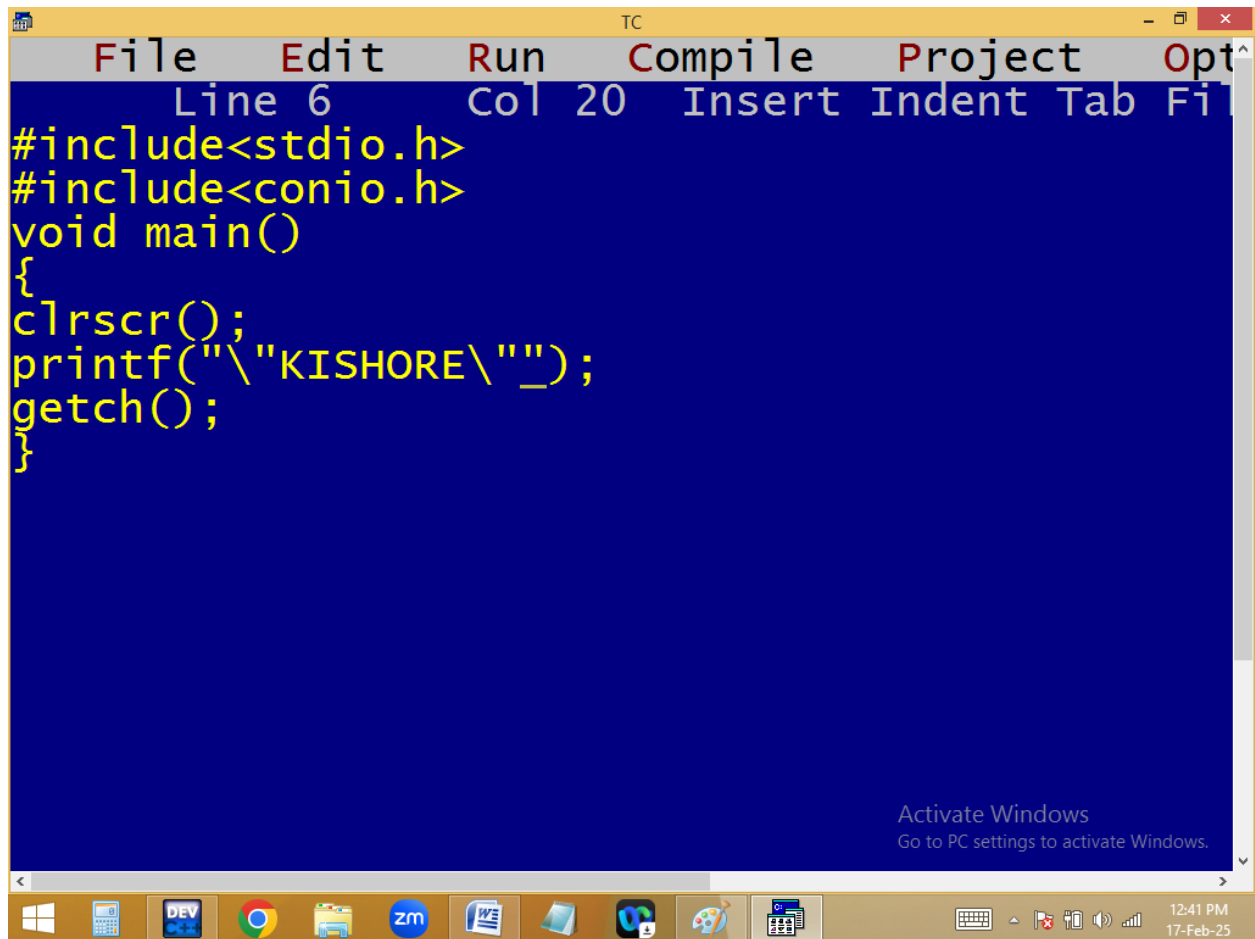


```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 56 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d,%d,%d,%d,%d,%d",'\0','\a','\b','\t','\n','a');
getch();
}
```

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12:39 PM 17-Feb-25

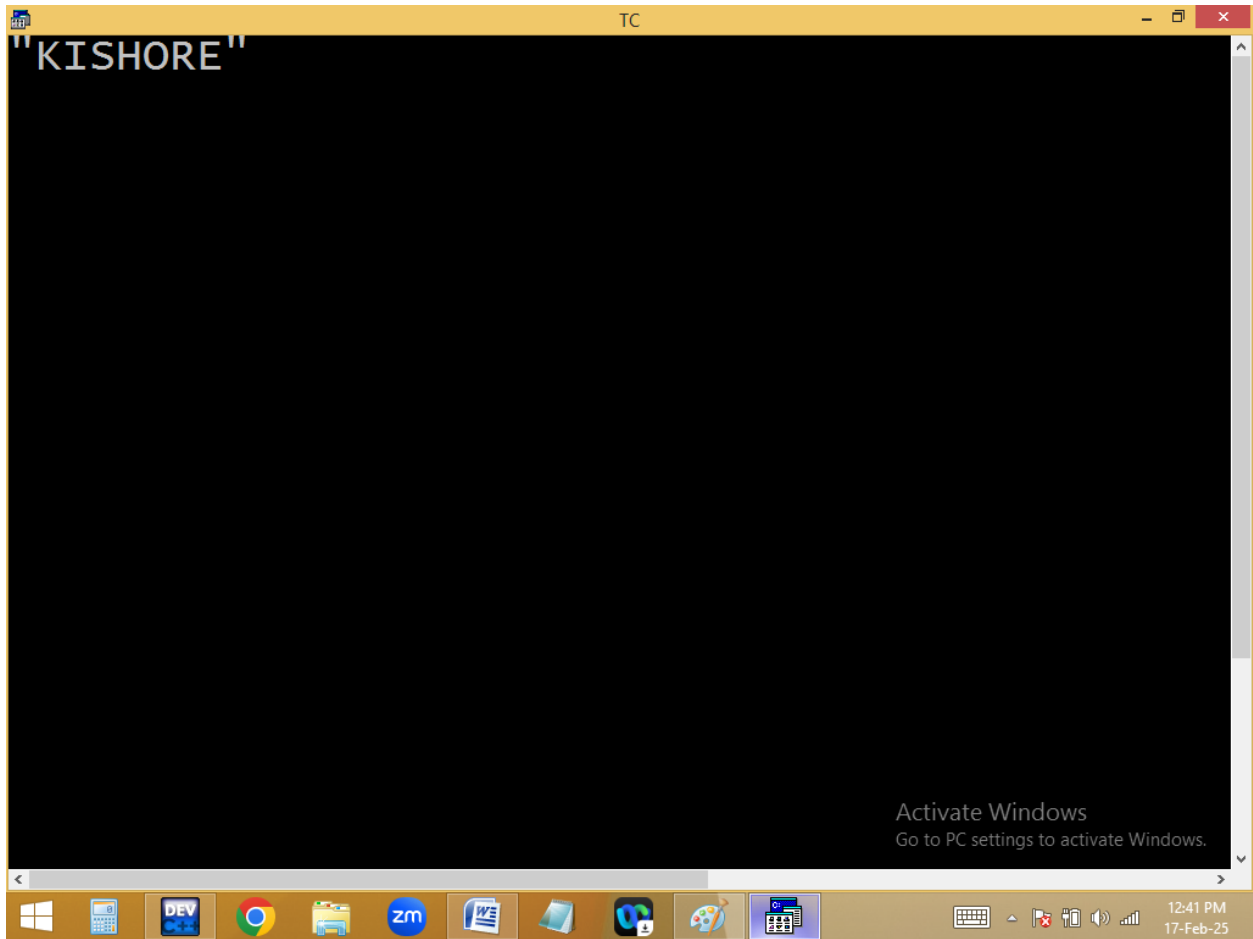




```
File Edit Run Compile Project Options
Line 6 Col 20 Insert Indent Tab File
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("\nKISHORE\'_");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

12:41 PM
17-Feb-25



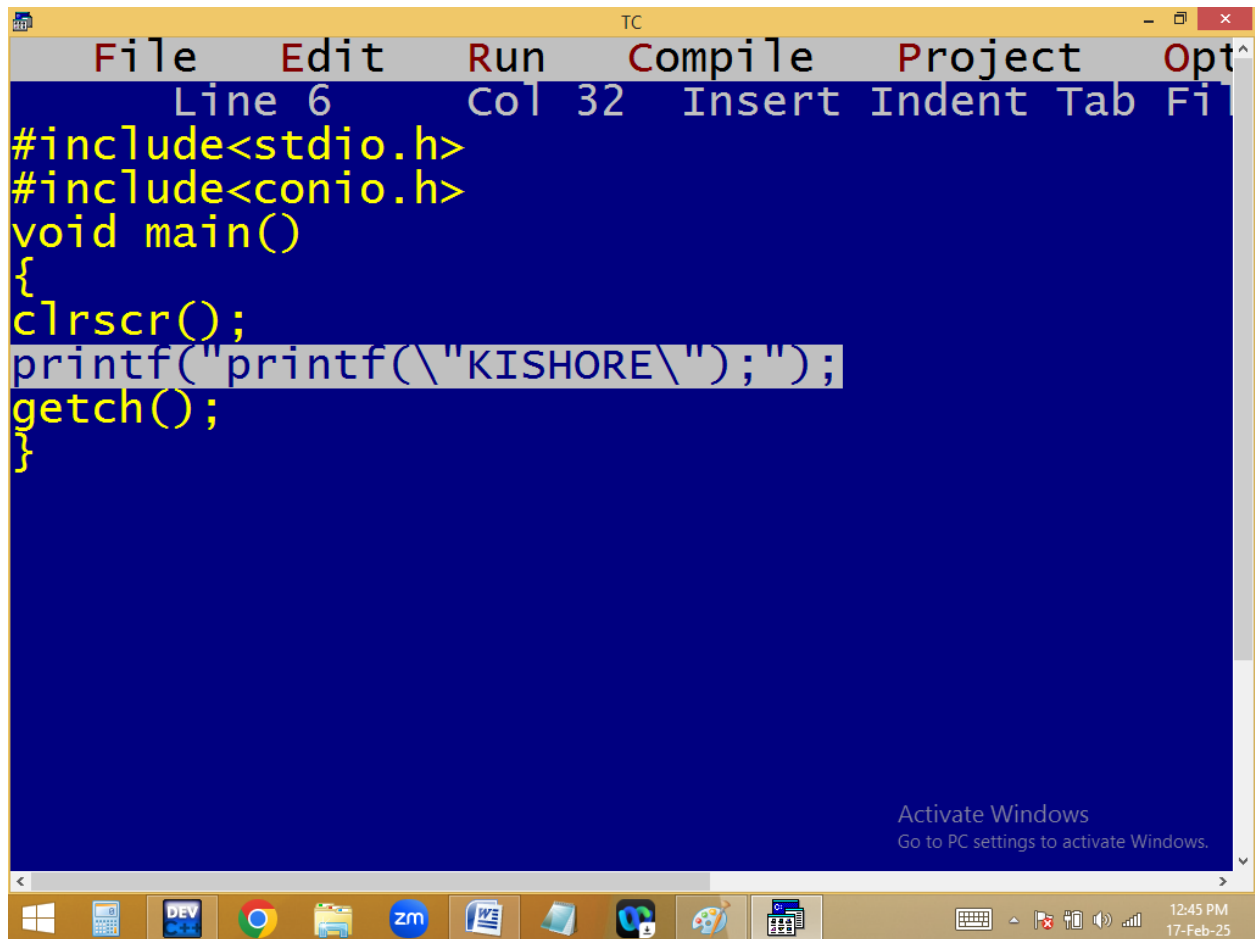
// - \

p("\KISHORE\");

\K - K

\X - X

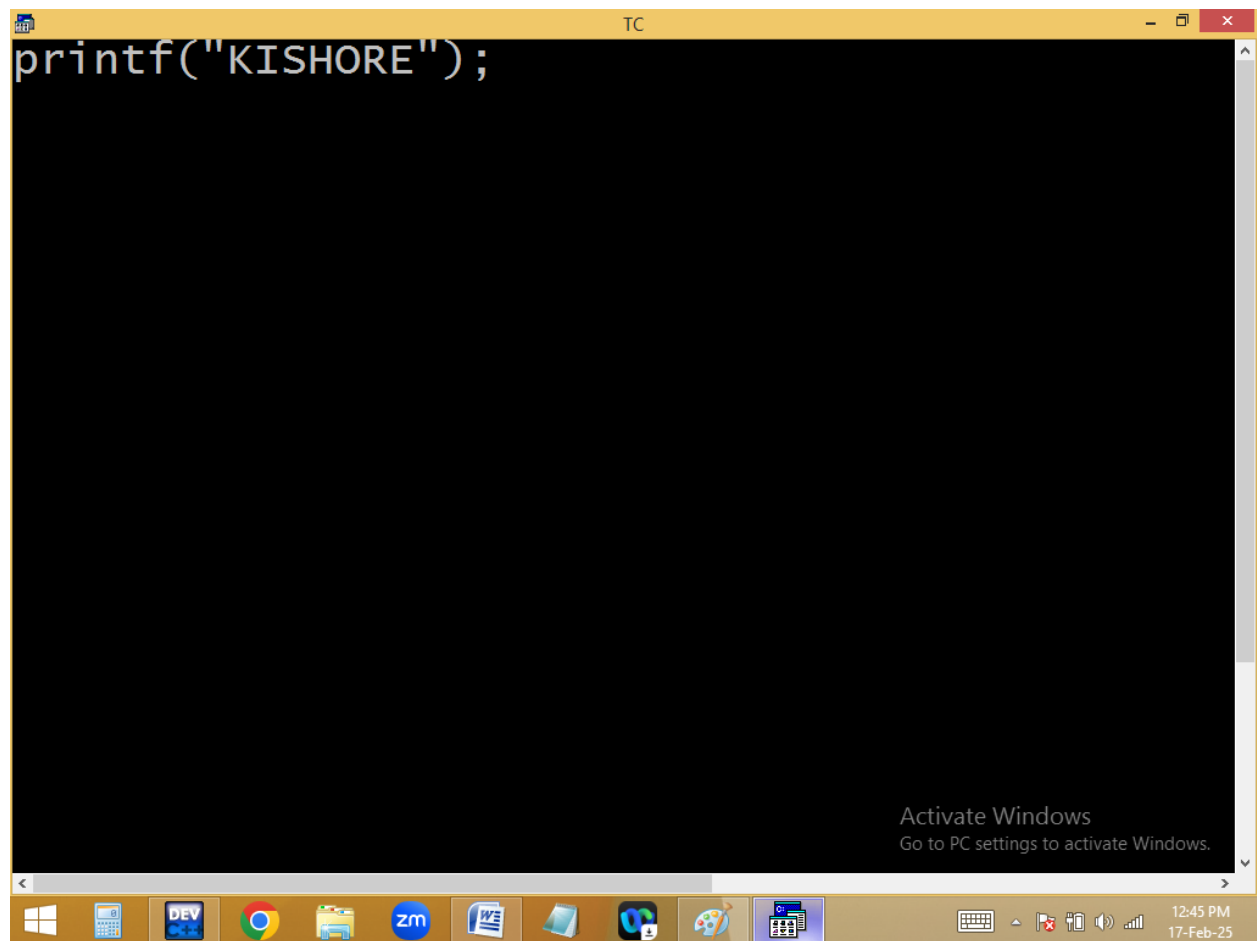
\ - \"



```
TC
File Edit Run Compile Project Options
Line 6 Col 32 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("printf(\"KISHORE\");");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

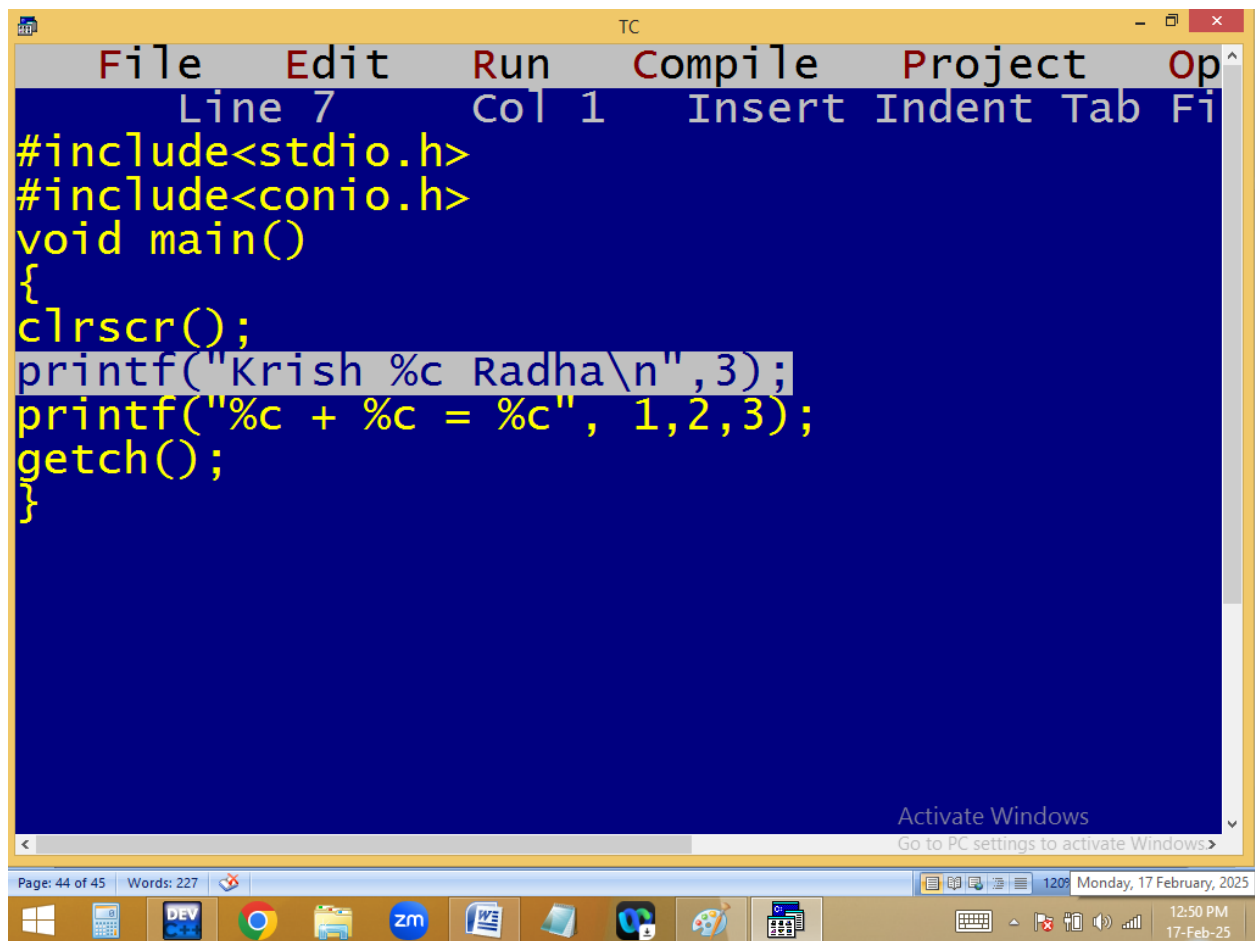
12:45 PM
17-Feb-25



```
printf("KISHORE");
```

Activate Windows
Go to PC settings to activate Windows.

`printf("printf(\"Kishore\");");`



```
File Edit Run Compile Project Op
Line 7 Col 1 Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Krish %c Radha\n",3);
printf("%c + %c = %c", 1,2,3);
getch();
}
```

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